

CCT College Dublin

Assessment Cover Page

Module Title:	BSc in Computing and IT
Assessment Title:	Proof of Concept Linux Virtual Network Project
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Operating Systems

Proof of Concept Linux Virtual Network Project

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Assignment Introduction

On this Report I'll be demonstrating my understanding on how to configure a Linux Machine by taking a new role in an ICT consultancy named Consult & Connect Ltd and providing consultancy services to a college called Dublin City College (DCC).

I created a virtualized network, with two servers (the web server and the web client server) and configured accordingly to what was asked. Through this assignment I installed the Apache web server and Lynx browser on web client (Part 1). After checking if both were pinging each other and other machines, I monitored the traffic using Wireshark.

Later on, DCC requested to change the Hostname (Part1B) and the IP address (Part 1C) permanently. It was also requested to install and configure a DHCP server to allocate the client address (Part 1D). SSH was used to transfer data securely between the two servers and using Putty to check that data (Part 2). For more safety it was requested a Security Configuration to the Uncomplicated Firewall to control the traffic (Part 3). To finish, it was created a SAMBA file server to allow Ubuntu and Windows computers to interact with each other (Part 4).

At the end I'll be completing some challenges (Part 5).

Installing the new Linux servers:

Ubuntuserver (The Web server)

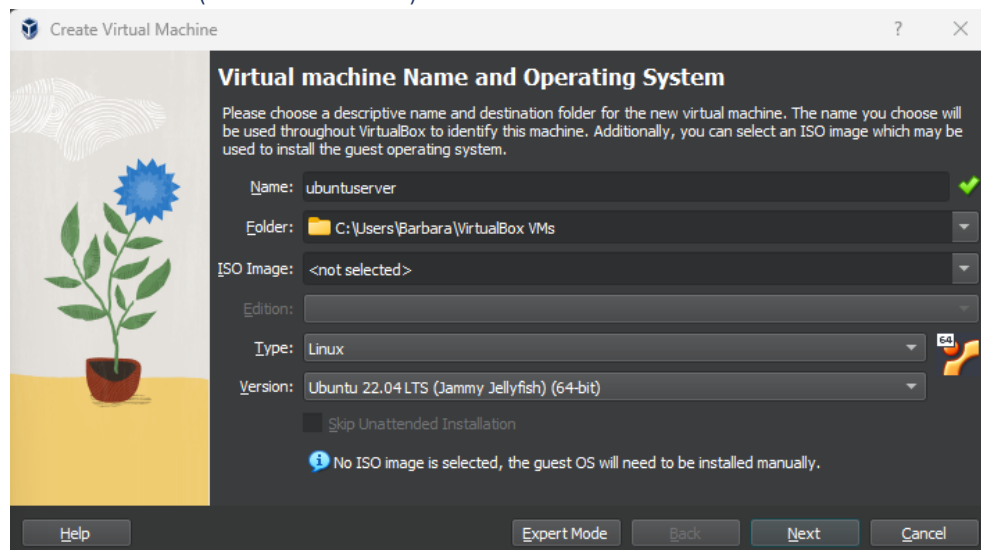


Figure 1 Creating a Linux Virtual Machine (Ubuntuserver)

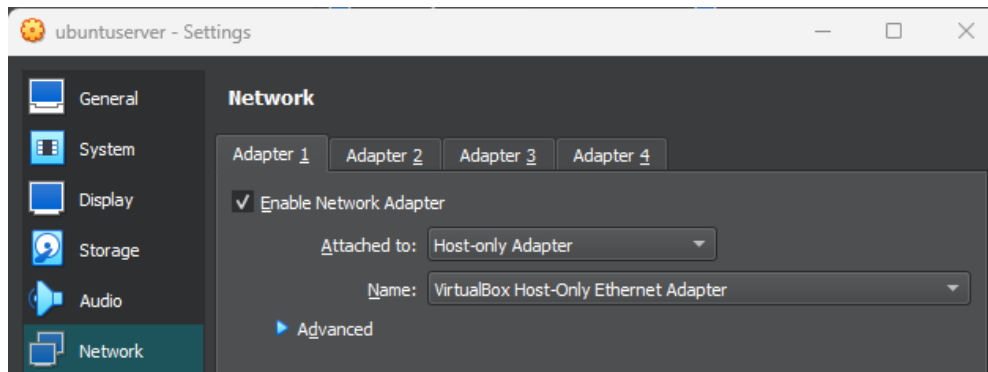


Figure 2 Setting up the Host-only Adaptor

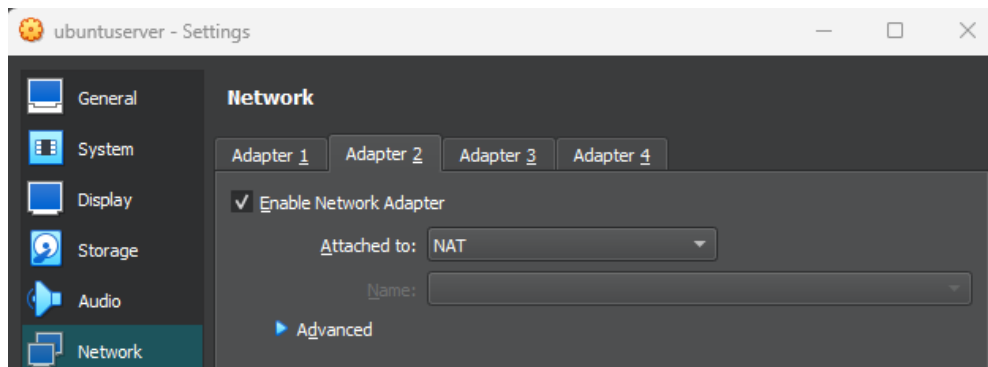


Figure 3 Setting up the NAT Adaptor

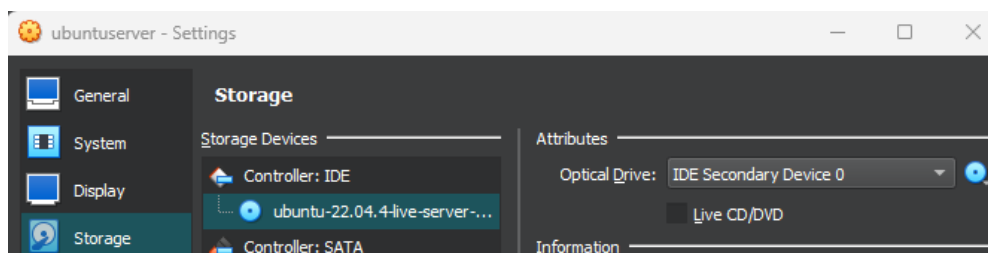


Figure 4 Adding the ubuntu iso image in Optical Drive

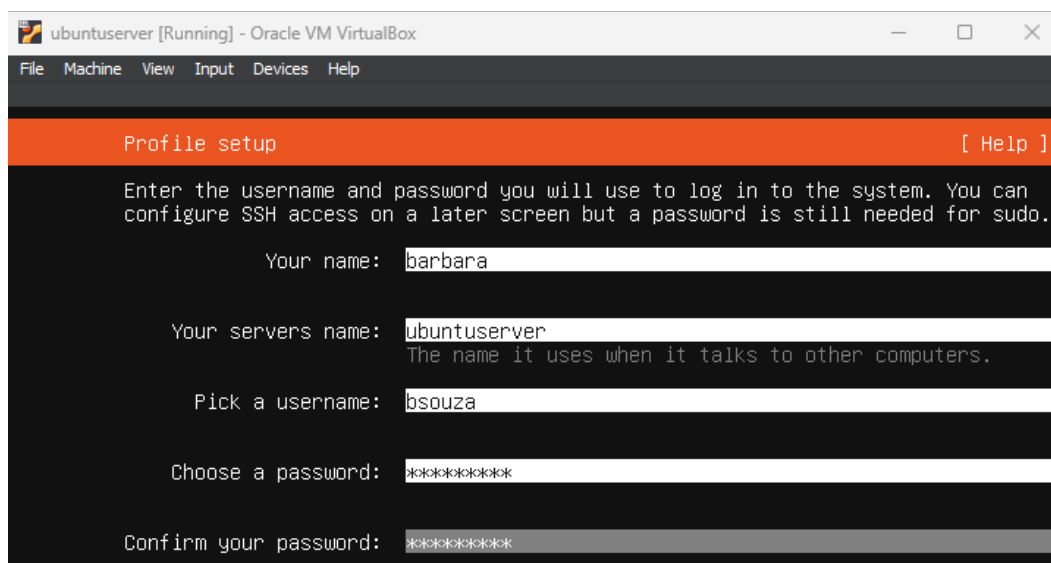


Figure 5 Linux Profile setup

Ubuntucient (The Web Client)

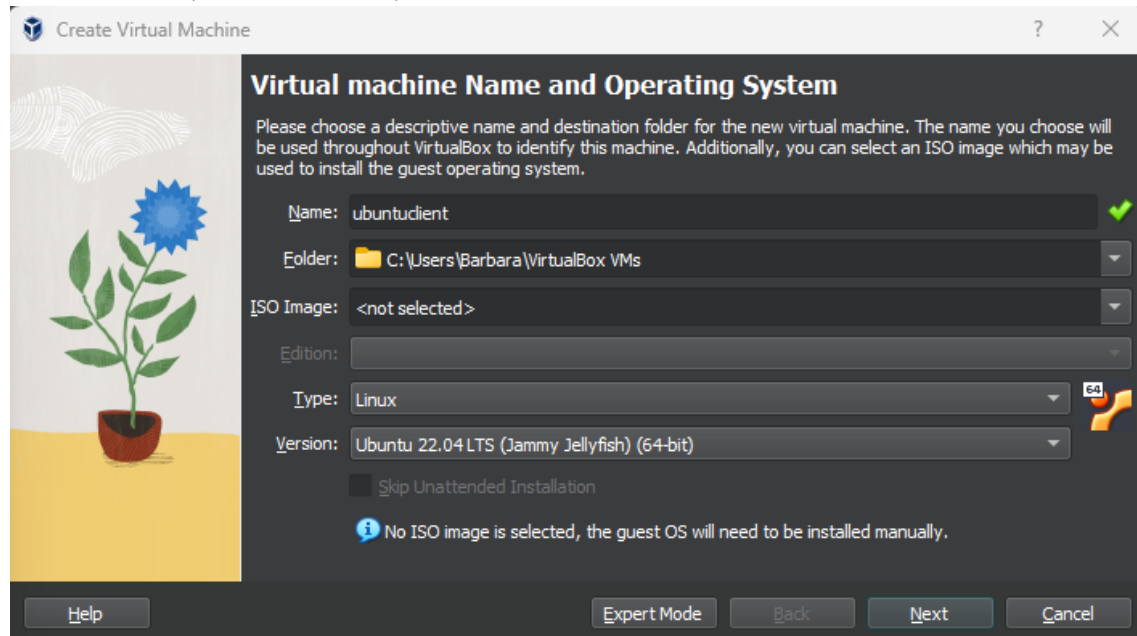


Figure 6 Creating Linux Virtual Machine (Ubuntucient)

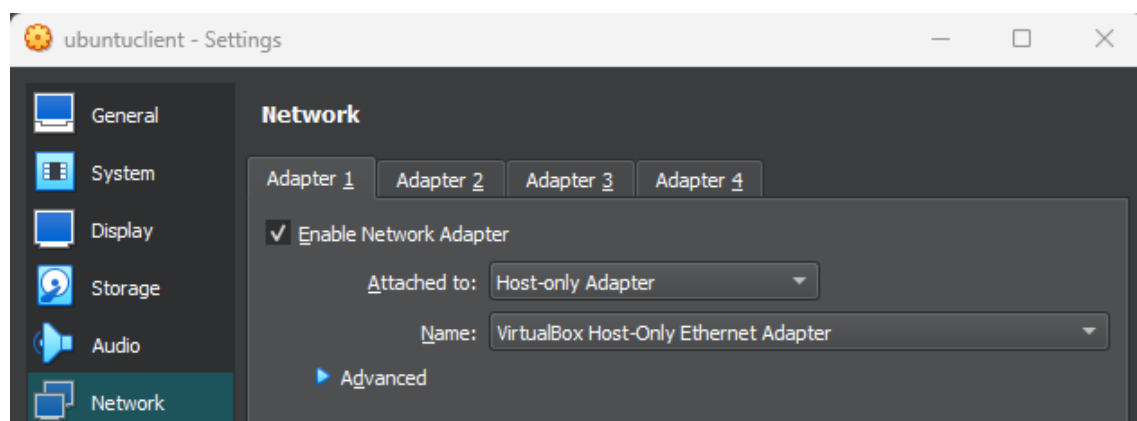


Figure 7 Setting up the Host-only Adaptor

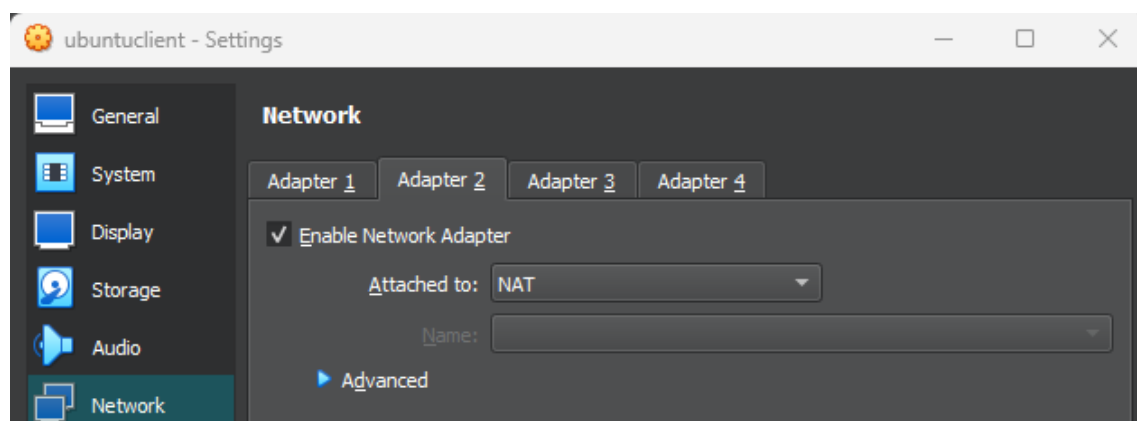


Figure 8 Setting up the NAT Adaptor

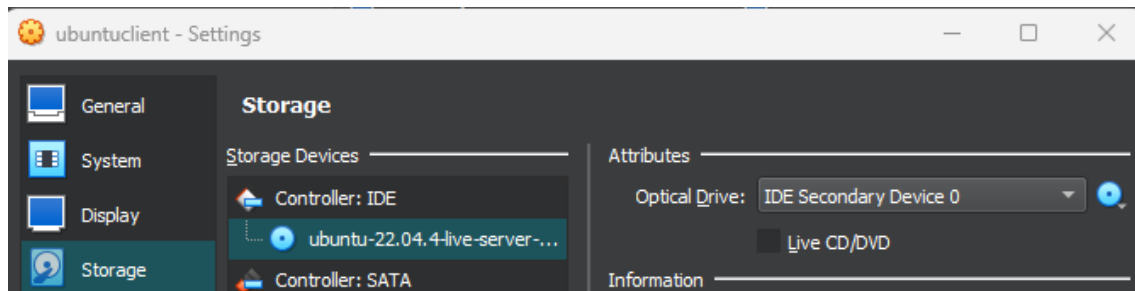


Figure 9 Adding the ubuntu iso image in Optical Drive

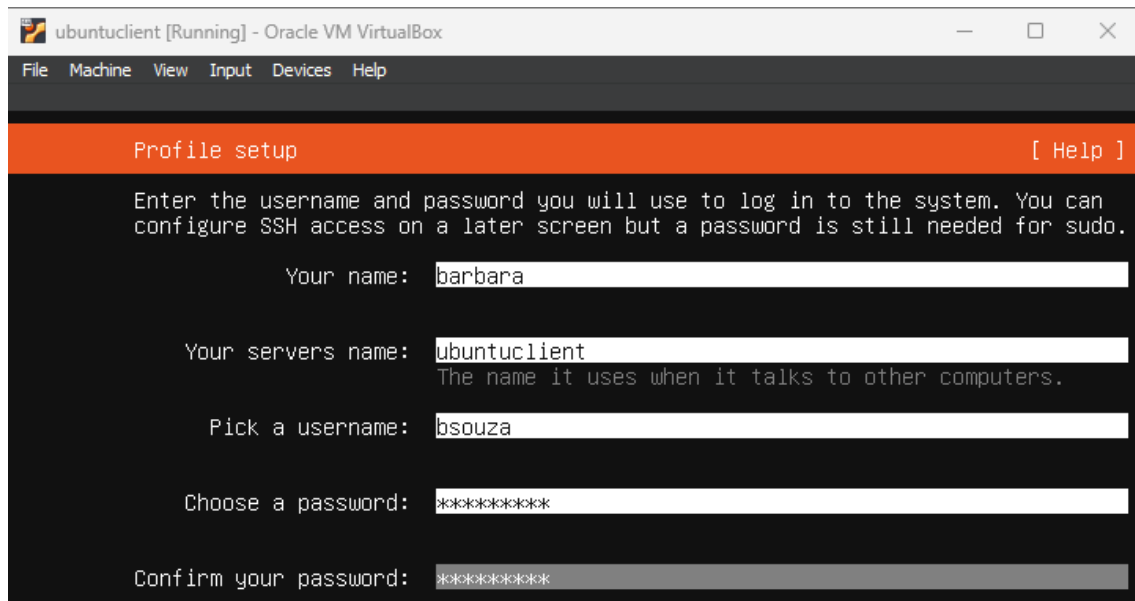
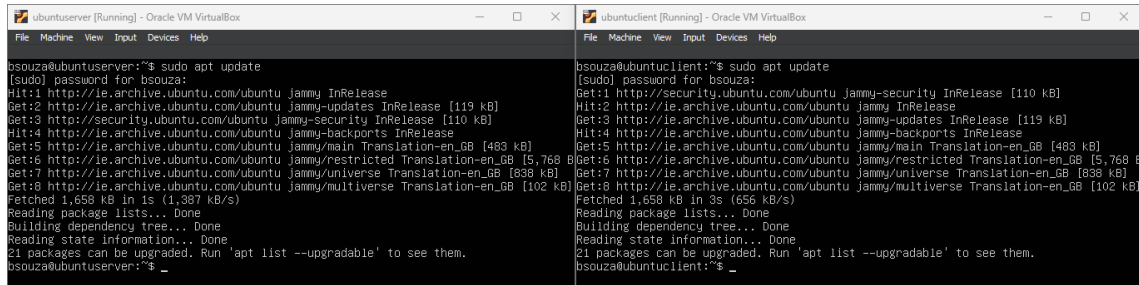


Figure 10 Linux Profile setup

Part 1 A: Virtual Linux Network and Web server

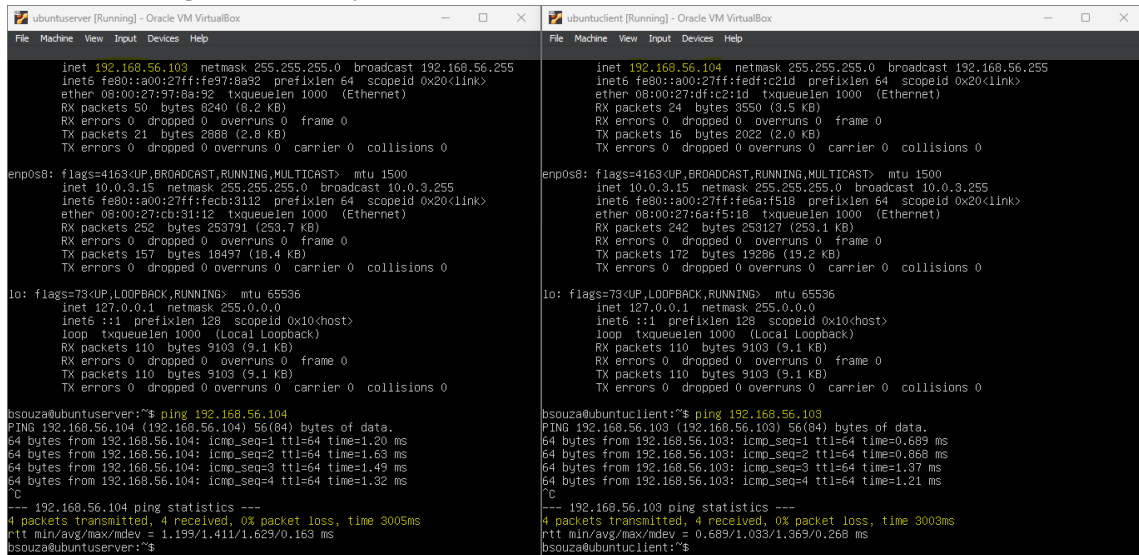
1. Updating the system



The image shows two side-by-side terminal windows. The left window is titled 'ubuntu:server [Running] - Oracle VM VirtualBox' and the right window is titled 'ubuntu:client [Running] - Oracle VM VirtualBox'. Both windows show the command 'sudo apt update' being executed. The output for both is identical, showing the process of updating package lists and fetching updates from various sources like 'http://ie.archive.ubuntu.com/ubuntu'. The output indicates that 21 packages can be upgraded and provides a list of them.

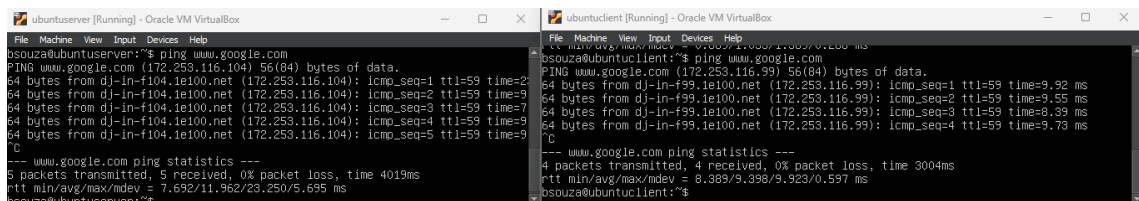
Figure 11 Updating both machines with “sudo apt update”

2. Testing Connectivity



The image shows two side-by-side terminal windows. The left window is titled 'ubuntu:server [Running] - Oracle VM VirtualBox' and the right window is titled 'ubuntu:client [Running] - Oracle VM VirtualBox'. Both windows show the output of the 'ifconfig' command for the 'eth0' interface, displaying IP addresses (192.168.56.103 and 192.168.56.104), netmasks, and broadcast addresses. Below this, the output of the 'ping' command is shown for both machines, indicating successful connectivity with 4 packets transmitted, 4 received, and 0% packet loss. The ping times are also displayed.

Figure 12 Both Machines Pinging each other



The image shows two side-by-side terminal windows. The left window is titled 'ubuntu:server [Running] - Oracle VM VirtualBox' and the right window is titled 'ubuntu:client [Running] - Oracle VM VirtualBox'. Both windows show the command 'ping www.google.com' being executed. The output for both is identical, showing the process of pinging google.com and displaying the results, including the IP address (172.253.116.104) and the ping times. The output indicates that 5 packets were transmitted, 5 received, and 0% packet loss, with a time of 4019ms.

Figure 13 Both Machines Pinging Google.com

```

ubuntu:server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
inet6 fe80::a00:27ff:fe97:8a92 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:97:8a:92 txqueuelen 1000 (Ethernet)
RX packets 99 bytes 12952 (12.9 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 59 bytes 6596 (6.5 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.0.3.15 netmask 255.255.255.0 broadcast 10.0.0.3.255
inet6 fe80::a00:27ff:feb3:3112 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:cb:31:12 txqueuelen 1000 (Ethernet)
RX packets 269 bytes 255544 (255.5 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 174 bytes 20008 (20.0 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 124 bytes 10615 (10.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 124 bytes 10615 (10.6 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bsouza@ubuntu:server~$ ping 192.168.1.110
PING 192.168.1.110 (192.168.1.110) 56(84) bytes of data:
64 bytes from 192.168.1.110: icmp_seq=1 ttl=127 time=1.28 ms
64 bytes from 192.168.1.110: icmp_seq=2 ttl=127 time=2.01 ms
64 bytes from 192.168.1.110: icmp_seq=3 ttl=127 time=2.07 ms
64 bytes from 192.168.1.110: icmp_seq=4 ttl=127 time=1.91 ms
^C
--- 192.168.1.110 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 1.283/1.816/2.065/0.313 ms
bsouza@ubuntu:server~$ _

Windows PowerShell
Sufixo DNS específico de conexão. . . . . :
Adaptador de Rede sem Fio Conexão Local* 10:
Estado da mídia. . . . . : mídia desconectada
Sufixo DNS específico de conexão. . . . . :
Adaptador de Rede sem Fio Wi-Fi:
Sufixo DNS específico de conexão. . . . . :
Endereço IPv6 de link local . . . . . : fe80::cc6:ddad:b2de:b41
9a10
Endereço IPv4. . . . . : 192.168.1.110
Máscara de Sub-rede . . . . . : 255.255.255.0
Gateway Padrão. . . . . : 192.168.1.1
Adaptador Ethernet Conexão de Rede Bluetooth:
Estado da mídia. . . . . : mídia desconectada
Sufixo DNS específico de conexão. . . . . :
PS C:\Users\Barbara> ping 192.168.56.103

Disparando 192.168.56.103 com 32 bytes de dados:
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64

Estatísticas do Ping para 192.168.56.103:
Pacotes: Enviados = 4, Recebidos = 4, Perdidos = 0 (0% de
perda),
Aproximar um número redondo de vezes em milissegundos:
Mínimo = 0ms, Máximo = 1ms, Média = 0ms
PS C:\Users\Barbara>

```

Figure 14 UbuntuServer pinging the host computer

```

Windows PowerShell
Adaptador Ethernet Conexão de Rede Bluetooth:
Estado da mídia. . . . . : mídia desconectada
Sufixo DNS específico de conexão. . . . . :
PS C:\Users\Barbara> ping 192.168.56.103

Disparando 192.168.56.103 com 32 bytes de dados:
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.103: bytes=32 tempo<1ms TTL=64

Estatísticas do Ping para 192.168.56.103:
Pacotes: Enviados = 4, Recebidos = 4, Perdidos = 0 (0% de
perda),
Aproximar um número redondo de vezes em milissegundos:
Mínimo = 0ms, Máximo = 1ms, Média = 0ms
PS C:\Users\Barbara>

ping 192.168.56.104

Disparando 192.168.56.104 com 32 bytes de dados:
Resposta de 192.168.56.104: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.104: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.104: bytes=32 tempo<1ms TTL=64
Resposta de 192.168.56.104: bytes=32 tempo<1ms TTL=64

Estatísticas do Ping para 192.168.56.104:
Pacotes: Enviados = 4, Recebidos = 4, Perdidos = 0 (0% de
perda),
Aproximar um número redondo de vezes em milissegundos:
Mínimo = 0ms, Máximo = 1ms, Média = 0ms
PS C:\Users\Barbara>

ubuntu:client [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
inet6 fe80::a00:27ff:fedf:c21d prefixlen 64 scopeid 0x20<link>
ether 08:00:27:df:c2:1d txqueuelen 1000 (Ethernet)
RX packets 97 bytes 11344 (11.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 35 bytes 4320 (4.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.0.3.15 netmask 255.255.255.0 broadcast 10.0.0.3.255
inet6 fe80::a00:27ff:fe6a:f518 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:6a:f5:18 txqueuelen 1000 (Ethernet)
RX packets 264 bytes 255330 (255.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 196 bytes 21388 (21.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 122 bytes 10399 (10.3 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 122 bytes 10399 (10.3 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

bsouza@ubuntu:client~$ ping 192.168.1.110
PING 192.168.1.110 (192.168.1.110) 56(84) bytes of data:
64 bytes from 192.168.1.110: icmp_seq=1 ttl=127 time=1.03 ms
64 bytes from 192.168.1.110: icmp_seq=2 ttl=127 time=1.53 ms
64 bytes from 192.168.1.110: icmp_seq=3 ttl=127 time=0.740 ms
64 bytes from 192.168.1.110: icmp_seq=4 ttl=127 time=1.10 ms
^C
--- 192.168.1.110 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 0.740/1.101/1.531/0.282 ms
bsouza@ubuntu:client~$ _

```

Figure 15 UbuntuClient pinging the host computer

3. Website Under Construction

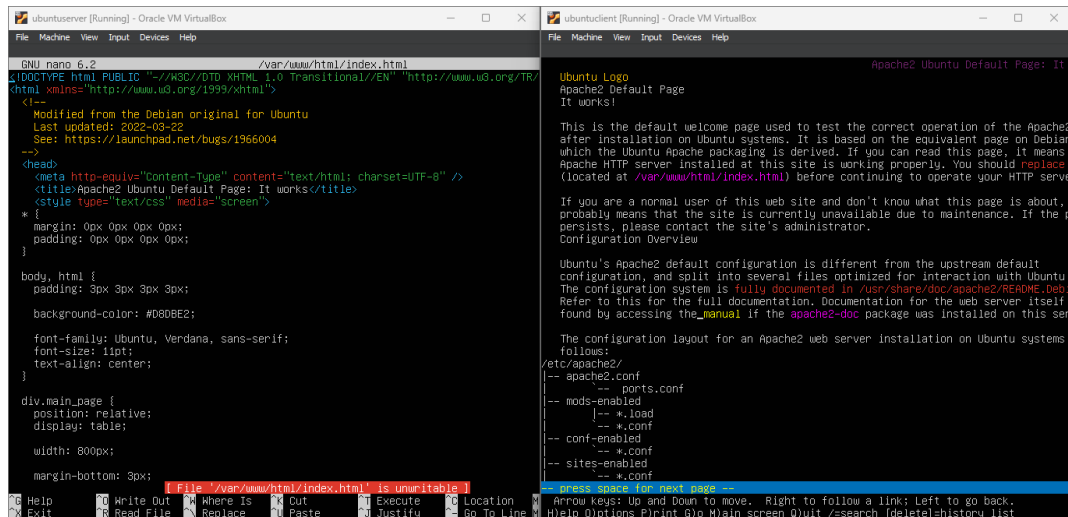


Figure 16 Website Under Construction opened in Ubuntuserver and and accessing it on Lynx browser on the Ubuntuclient

4. DCC on Windows host Operating System

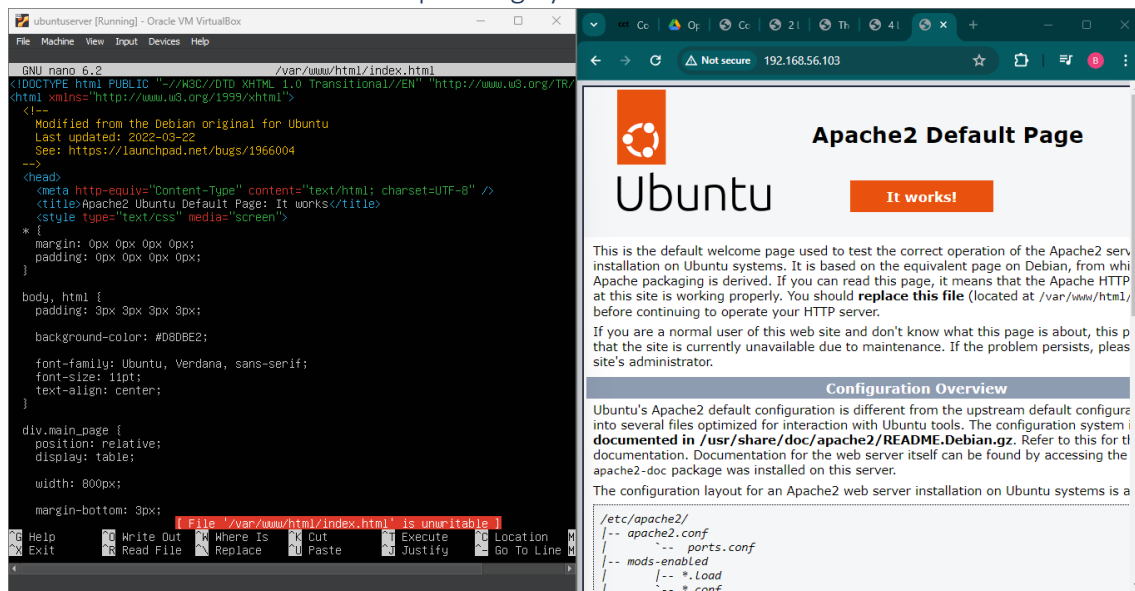


Figure 17 Website Under Construction opened in Ubuntuserver and and accessing it on my host machine

5. Wireshark ICMP Traffic

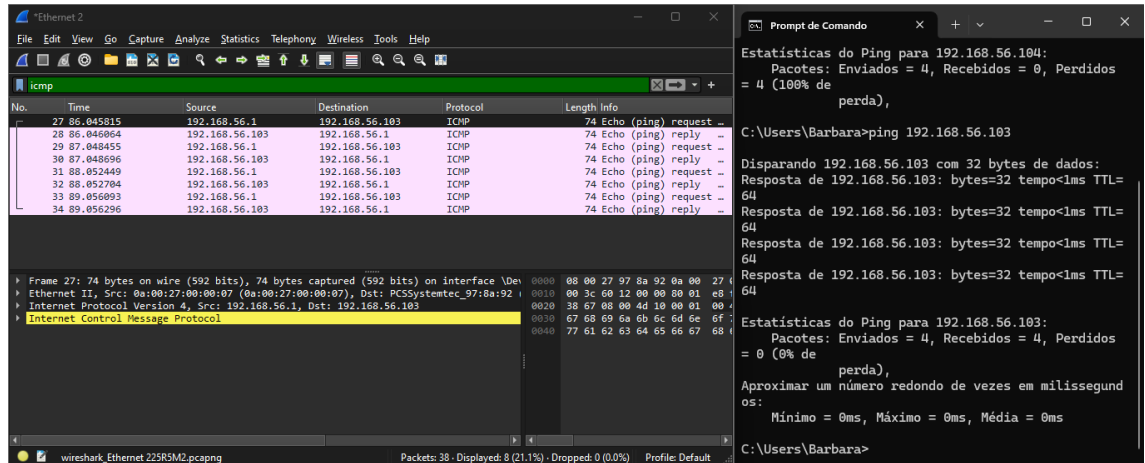


Figure 18 Monitoring the ICMP Traffic between my browser on my host operating system and the Ubuntu server with Wireshark

6. Wireshark TCP/IP 3-Way handshake

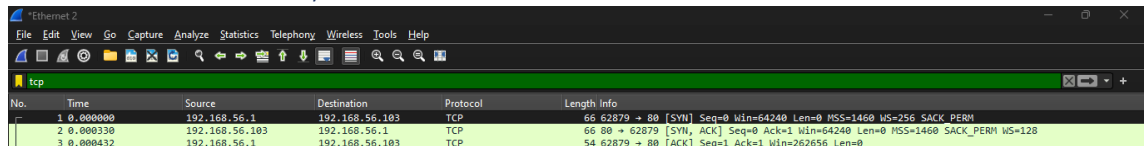


Figure 19 Using Wireshark to capture the TCP/IP 3-way handshake while the HTTP contents are being transferred from Ubuntu server to the host machine

Part 1B: HOSTNAME MANAGEMENT - Renaming the linux servers:

Web-server-059

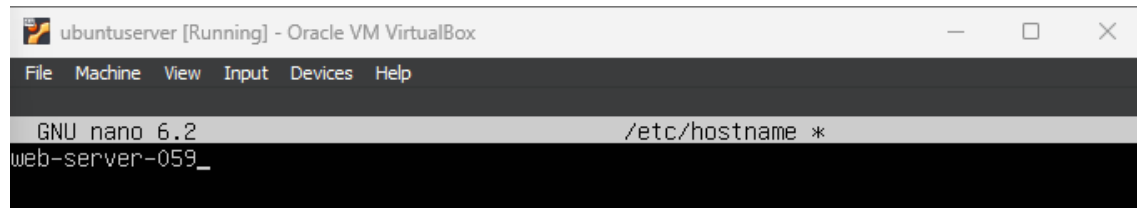


Figure 20 Changing the Hostname (Ubuntuserver) to web-server-059 permanently on the file /etc/hostname

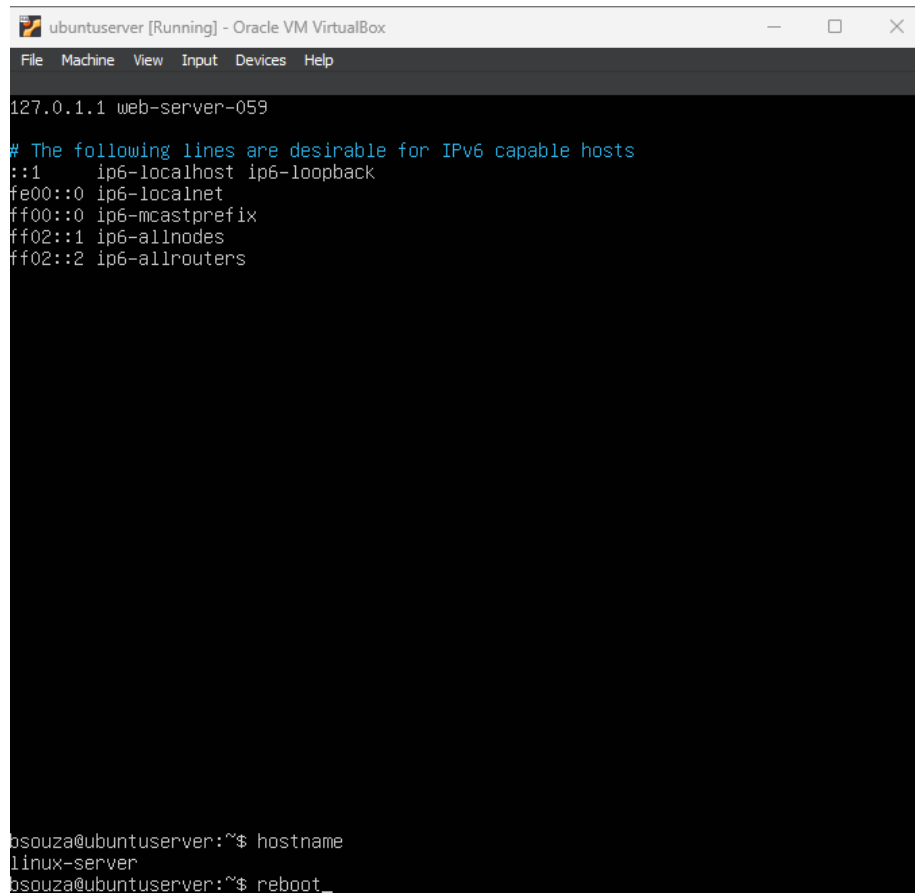


Figure 21 Changing the Hostname (Ubuntuserver) to web-server-059 permanently on the file /etc/hosts then Rebooting the machine

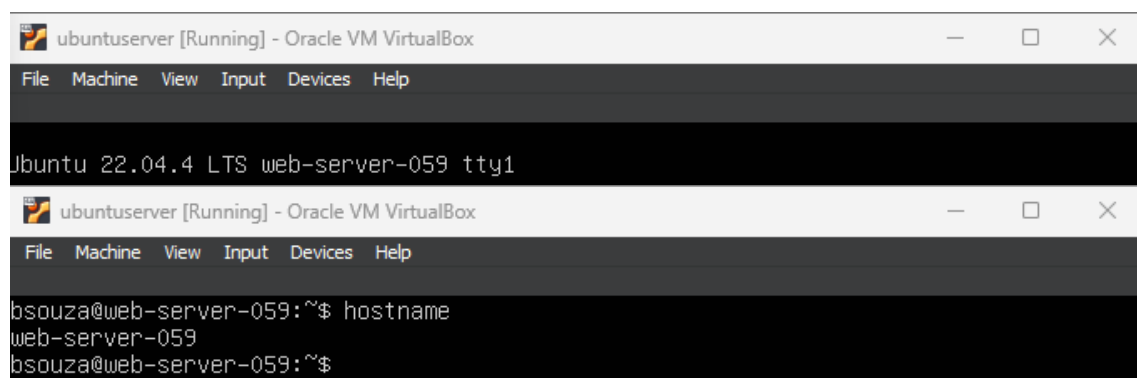
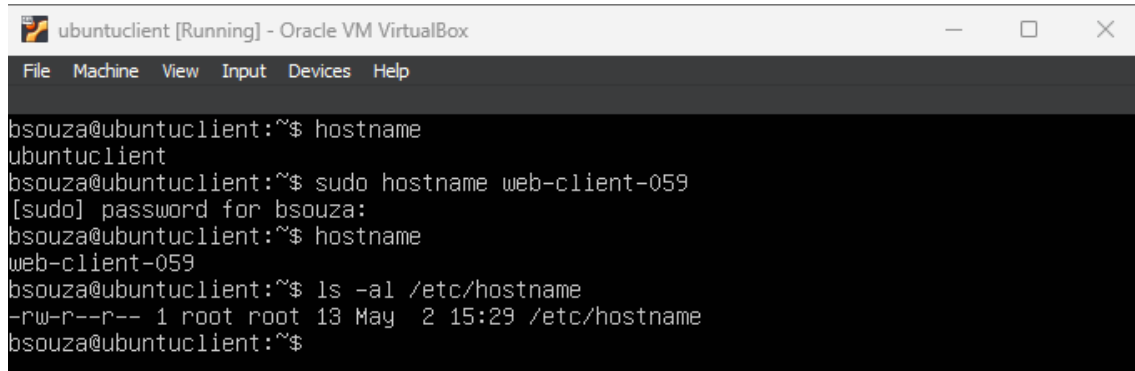


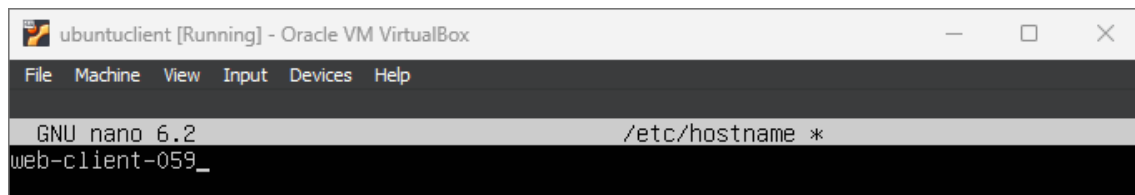
Figure 22 Hostname after Reboot

Web-client-059

A terminal window titled 'ubuntucient [Running] - Oracle VM VirtualBox' with a menu bar (File, Machine, View, Input, Devices, Help). The terminal shows a user 'bsouza' at 'ubuntucient' running 'hostname', then 'sudo hostname web-client-059' (prompting for password), then 'hostname' again showing 'web-client-059', and finally 'ls -al /etc/hostname' showing the file permissions and timestamp.

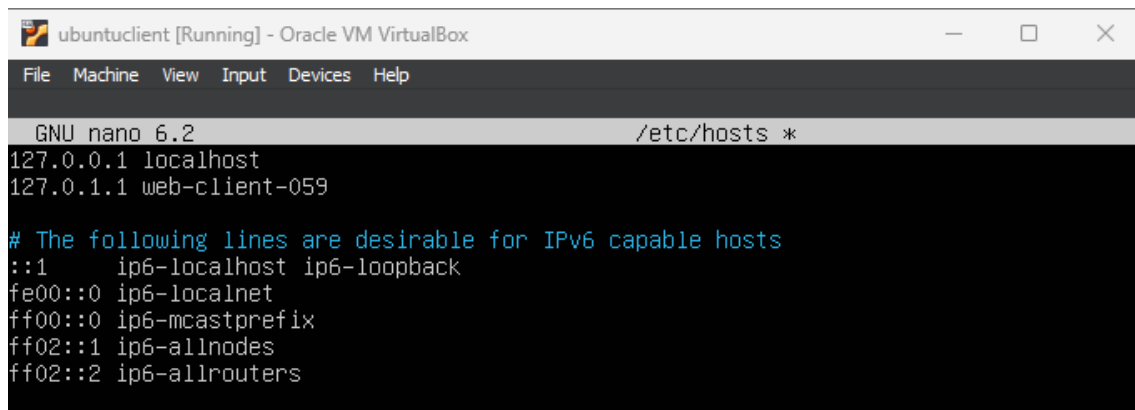
```
bsouza@ubuntucient:~$ hostname
ubuntucient
bsouza@ubuntucient:~$ sudo hostname web-client-059
[sudo] password for bsouza:
bsouza@ubuntucient:~$ hostname
web-client-059
bsouza@ubuntucient:~$ ls -al /etc/hostname
-rw-r--r-- 1 root root 13 May  2 15:29 /etc/hostname
bsouza@ubuntucient:~$
```

Figure 23 Changing hostname with "sudo hostname wbe-client-059"

A terminal window titled 'ubuntucient [Running] - Oracle VM VirtualBox' with a menu bar. It shows the GNU nano 6.2 editor editing the file '/etc/hostname *'. The current line is 'web-client-059_'.

```
GNU nano 6.2 /etc/hostname *
web-client-059_
```

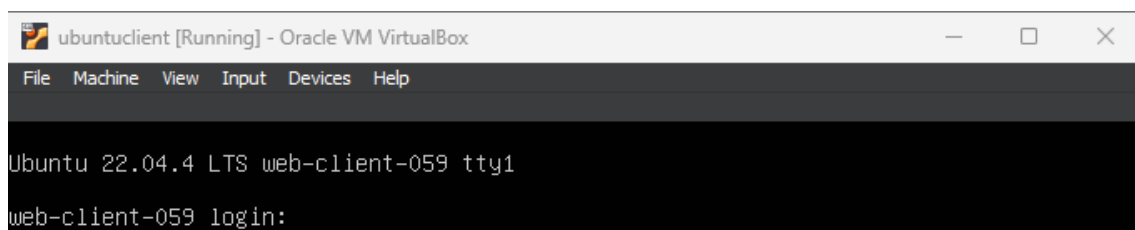
Figure 24 Changing the Hostname (Ubuntucient) to web-client-059 permanently on the file /etc/hostname

A terminal window titled 'ubuntucient [Running] - Oracle VM VirtualBox' with a menu bar. It shows the GNU nano 6.2 editor editing the file '/etc/hosts *'. The content includes IP addresses for localhost and web-client-059, followed by IPv6 addresses and their corresponding hostnames.

```
GNU nano 6.2 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 web-client-059

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Figure 25 Changing the Hostname (Ubuntucient) to web-client-059 permanently on the file /etc/hosts then Rebooting the machine

A terminal window titled 'ubuntucient [Running] - Oracle VM VirtualBox' with a menu bar. It shows the Ubuntu 22.04.4 LTS login screen for 'web-client-059' on 'tty1'.

```
Ubuntu 22.04.4 LTS web-client-059 tty1
web-client-059 login:
```

Figure 26 New Hostname after Rebooting

Part 1C: IP ADDRESS MANAGEMENT

Changing the IP Address permanently to 192.168.56.100/24

```

total 2097232
drwxr-xr-x 20 root root      4096 May  2 15:19 .
drwxr-xr-x 20 root root      4096 May  2 15:19 ..
lrwxrwxrwx  1 root root         7 Feb 16 18:37 bin -> usr/bin
drwxr-xr-x  4 root root      4096 May  2 15:23 boot
dr-xr-xr-x  2 root root      4096 Feb 16 23:52 cdrom
drwxr-xr-x 20 root root      4080 May  9 08:25 dev
drwxr-xr-x 100 root root      4096 May  3 16:55 etc
drwxr-xr-x  3 root root      4096 May  2 15:26 home
lrwxrwxrwx  1 root root         7 Feb 16 18:37 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Feb 16 18:37 lib32 -> usr/lib32
lrwxrwxrwx  1 root root         9 Feb 16 18:37 lib64 -> usr/lib64
lrwxrwxrwx  1 root root        10 Feb 16 18:37 libx32 -> usr/libx32
drwx-----  2 root root    16384 May  2 15:12 lost+found
drwxr-xr-x  2 root root      4096 Feb 16 18:37 media
drwxr-xr-x  2 root root      4096 Feb 16 18:37 mnt
drwxr-xr-x  2 root root      4096 Feb 16 18:37 opt
dr-xr-xr-x 185 root root         0 May  3 15:22 proc
drwx-----  5 root root      4096 May  3 15:16 root
drwxr-xr-x 34 root root      980 May  9 10:28 run
lrwxrwxrwx  1 root root         8 Feb 16 18:37 sbin -> usr/sbin
drwxr-xr-x  6 root root      4096 Feb 16 18:52 snap
drwxr-xr-x  2 root root      4096 Feb 16 18:37 srv
-rw-----  1 root root 2147483648 May  2 15:19 swap.img
dr-xr-xr-x 13 root root         0 May  3 15:22 sys
drwxrwxrwt 15 root root      4096 May  9 10:28 tmp
drwxr-xr-x 14 root root      4096 Feb 16 18:37 usr
drwxr-xr-x 14 root root      4096 May  2 16:13 var
bsouza@web-server-059:/$ /etc
-bash: /etc: Is a directory
bsouza@web-server-059:/$ cd /etc/netplan
bsouza@web-server-059:/etc/netplan$ ls -al
total 12
drwxr-xr-x  2 root root 4096 May  2 15:13 .
drwxr-xr-x 100 root root 4096 May  3 16:55 ..
-rw-----  1 root root  147 May  2 15:13 00-installer-config.yaml
bsouza@web-server-059:/etc/netplan$ _

```

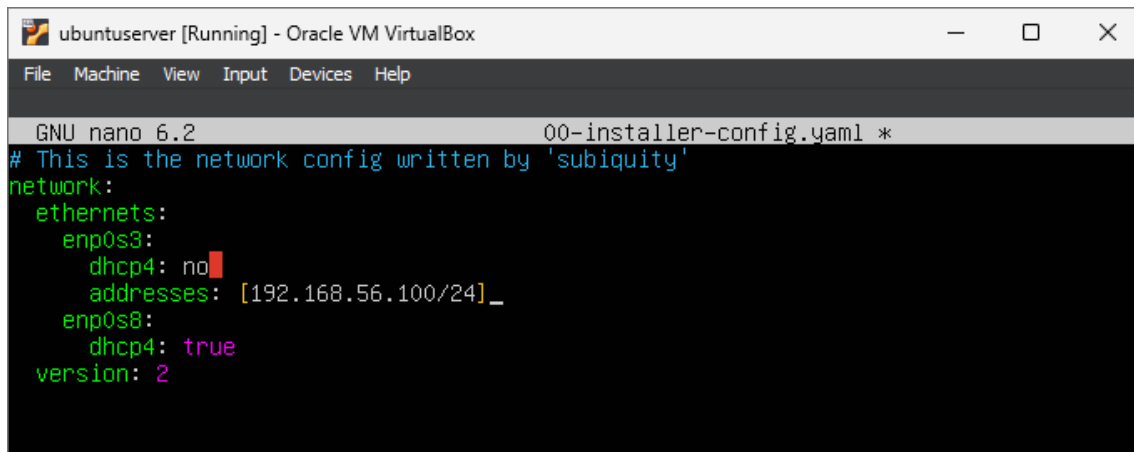
Figure 27 Opening the "cd /etc/netplan" folder

```

bsouza@web-server-059:/$ cd /etc/netplan
bsouza@web-server-059:/etc/netplan$ ls -al
total 12
drwxr-xr-x  2 root root 4096 May  2 15:13 .
drwxr-xr-x 100 root root 4096 May  3 16:55 ..
-rw-----  1 root root  147 May  2 15:13 00-installer-config.yaml
bsouza@web-server-059:/etc/netplan$ sudo nano 00-installer-config.yaml
[sudo] password for bsouza:

```

Figure 28 Opening the file with "sudo nano 00-installer-config.yaml"

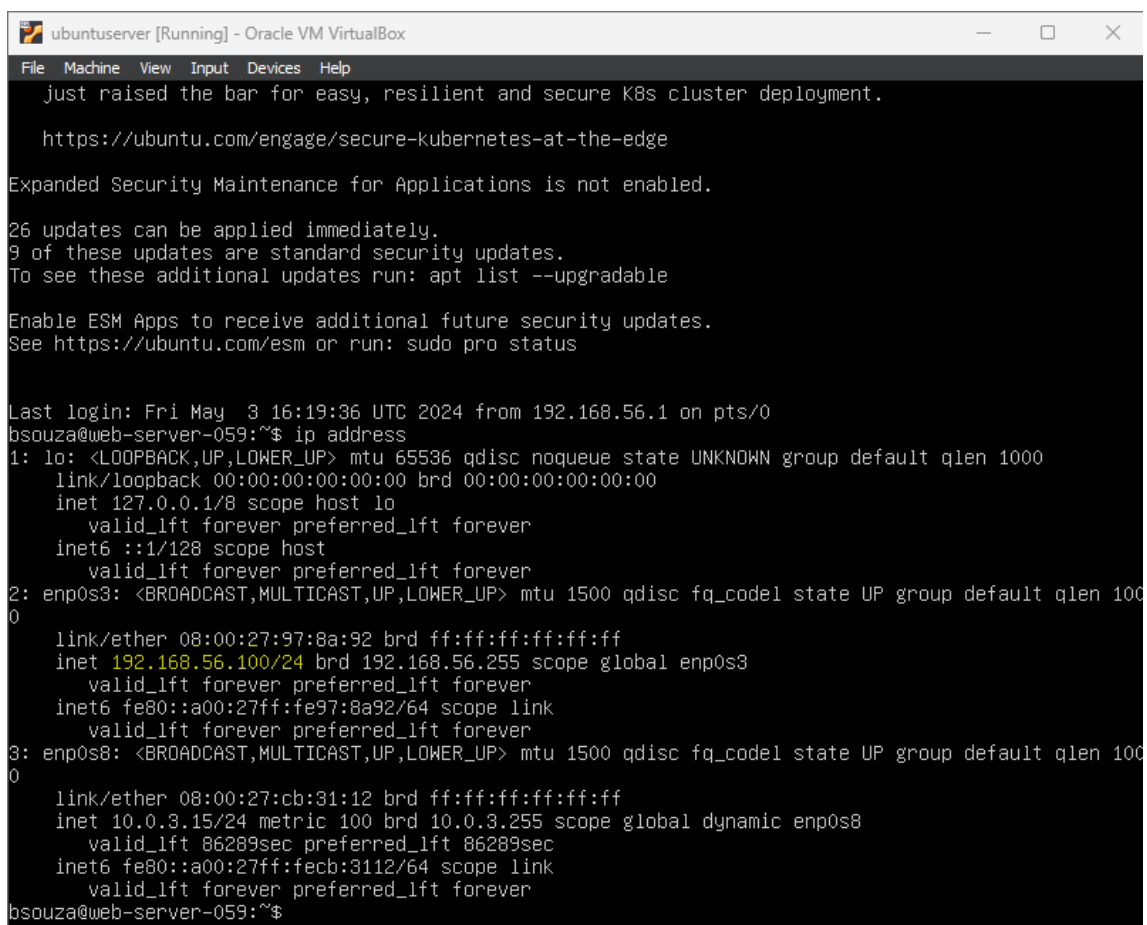


```

ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 6.2                                00-installer-config.yaml *
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.56.100/24]_
    enp0s8:
      dhcp4: true
  version: 2

```

Figure 29 Configuring the IP Address 192.168.56.100/24 to be permanent



```

ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

26 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 3 16:19:36 UTC 2024 from 192.168.56.1 on pts/0
bsouza@web-server-059:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:97:8a:92 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.100/24 brd 192.168.56.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe97:8a92/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cb:31:12 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 metric 100 brd 10.0.3.255 scope global dynamic enp0s8
        valid_lft 86289sec preferred_lft 86289sec
    inet6 fe80::a00:27ff:fe3b:3112/64 scope link
        valid_lft forever preferred_lft forever
bsouza@web-server-059:~$

```

Figure 30 After Rebooting the IP Address stayed as 192.168.56.100

Part 1D: IP ADDRESS MANAGEMENT WITH DHCP - Cancelled

PART 2: Using SSH for secure transfer of data between the two servers

7. Checking, enabling and allowing the ufw

```

bsouza@web-server-059:~$ sudo ufw status
[sudo] password for bsouza:
Status: inactive
bsouza@web-server-059:~$ sudo ufw enable
Firewall is active and enabled on system startup
bsouza@web-server-059:~$ sudo ufw status
Status: active
bsouza@web-server-059:~$ sudo ufw allow ssh
sudo: allow: command not found
bsouza@web-server-059:~$ sudo ufw allow ssh
Rule added (v6)
bsouza@web-server-059:~$ sudo ufw allow 22
Rule added
Rule added (v6)
bsouza@web-server-059:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)

```

Figure 31 Checking, enabling and allowing the ufw. Then, logging into ubuntu server using Putty.

- Checking ufw status with “sudo ufw status”

The ufw was inactive

- Enabling the ufw with “sudo ufw enable”
- Checking ufw status again

The ufw was now active

- Allowing the SSH traffic and port 22 with “sudo ufw allow ssh/22”
- Then Checking the ufw status again

The ufw still active and allowing SSH and 22 traffic.

8. Wireshark monitoring the encrypted traffic (SSH)

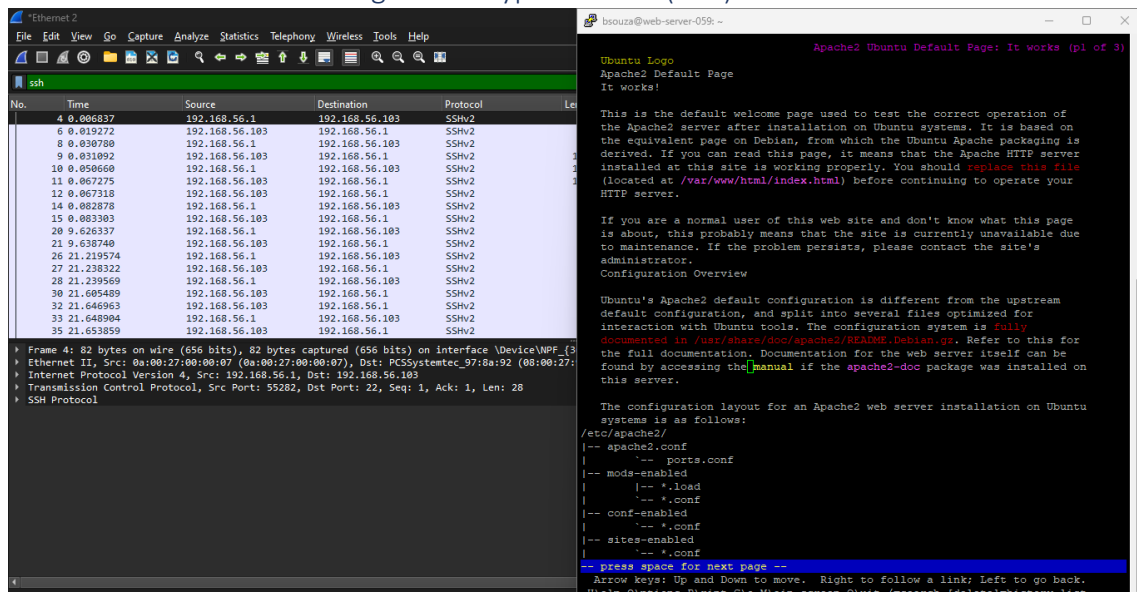


Figure 32 Wireshark monitoring the communication (Encrypted SSH) between the client operating system and the ubuntu server

PART 3: Security configuration #1:

```

Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22 DENY Anywhere
80/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)
80 (v6) DENY Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)

bsouza@web-server-059:~$ sudo ufw deny 22
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw deny ssh
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw status
Status: active

To Action From
--
22/tcp DENY Anywhere
22 DENY Anywhere
80 DENY Anywhere
80/tcp DENY Anywhere
22/tcp (v6) DENY Anywhere (v6)
22 (v6) DENY Anywhere (v6)
80 (v6) DENY Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)

bsouza@web-server-059:~$

```

Figure 33 Using the UFW to block the http traffic. Not being able to access it on Putty

```

80 ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) DENY Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

bsouza@web-server-059:~$ sudo ufw deny 22
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw deny 80
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw deny http
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw allow 22
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw allow ssh
Rule updated
Rule updated (v6)
bsouza@web-server-059:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22 ALLOW Anywhere
80 DENY Anywhere
80/tcp DENY Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
22 (v6) ALLOW Anywhere (v6)
80 (v6) DENY Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)

bsouza@web-server-059:~$

```

Ubuntu Logo Apache2 Ubuntu Default Page: It works (pl of 3)

Apache2 Default Page
It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```

/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
-- press space for next page --

```

Figure 34 Using the UFW to allow the http traffic. Accessing it on Putty normally.

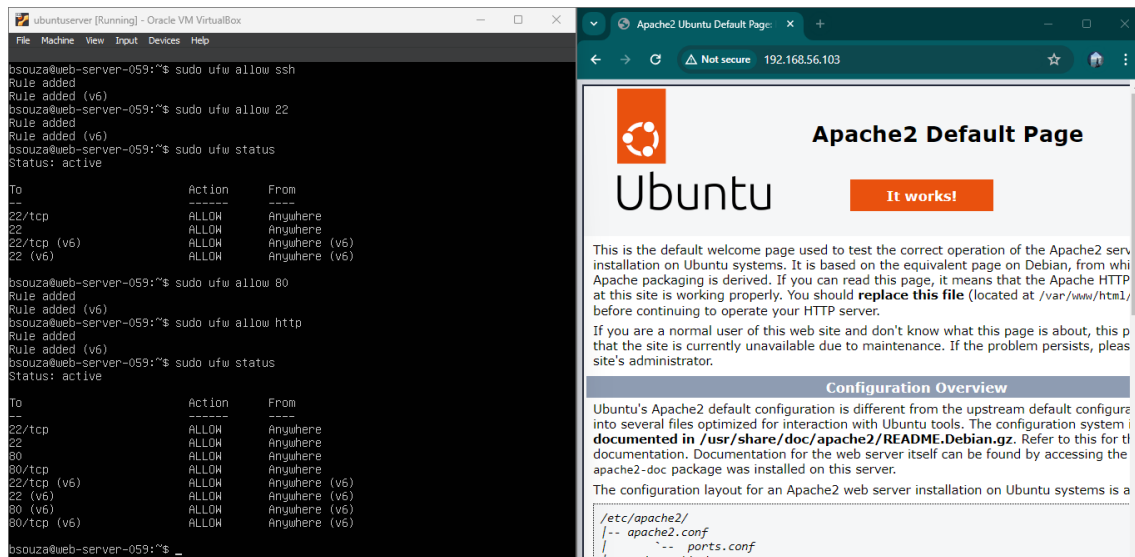


Figure 35 Using the UFW to allow SSH traffic. Accessing it on the host machine normally.

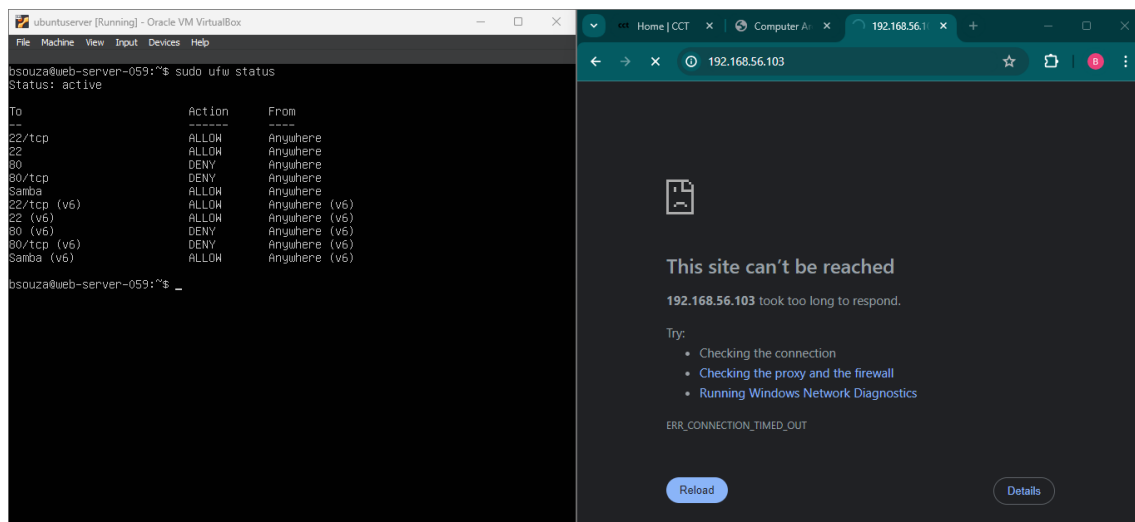
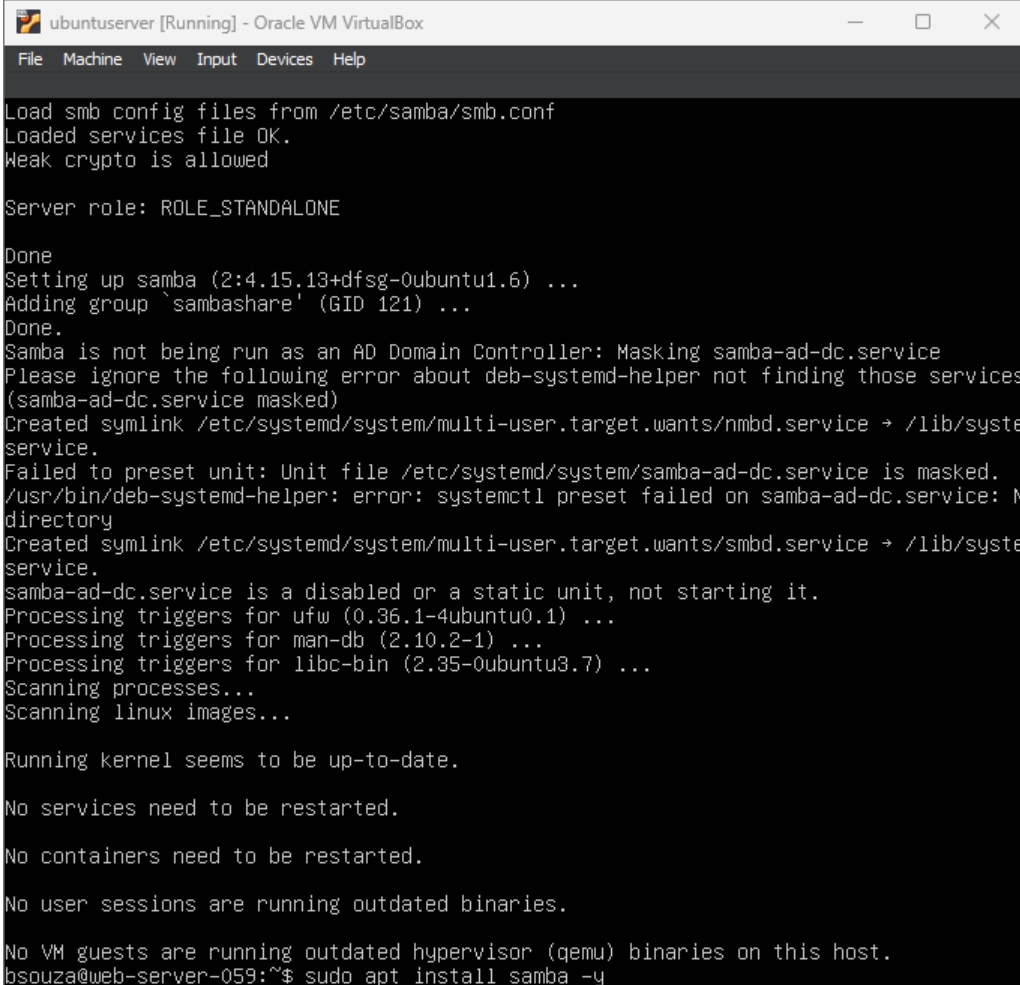


Figure 36 Using the UFW to deny SSH traffic. Not being able to access it on the host machine.

PART 4: The SAMBA file server

Configuring the Samba Server in Ubuntu Linux.



```

ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed

Server role: ROLE_STANDALONE

Done
Setting up samba (2:4.15.13+dfsg-0ubuntu1.6) ...
Adding group `sambashare' (GID 121) ...
Done.
Samba is not being run as an AD Domain Controller: Masking samba-ad-dc.service
Please ignore the following error about deb-systemd-helper not finding those services
(samba-ad-dc.service masked)
Created symlink /etc/systemd/system/multi-user.target.wants/nmbd.service → /lib/systemd/system/nmbd.service.
Failed to preset unit: Unit file /etc/systemd/system/samba-ad-dc.service is masked.
/usr/bin/deb-systemd-helper: error: systemctl preset failed on samba-ad-dc.service: No such directory
Created symlink /etc/systemd/system/multi-user.target.wants/smbd.service → /lib/systemd/system/smbd.service.
samba-ad-dc.service is a disabled or a static unit, not starting it.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

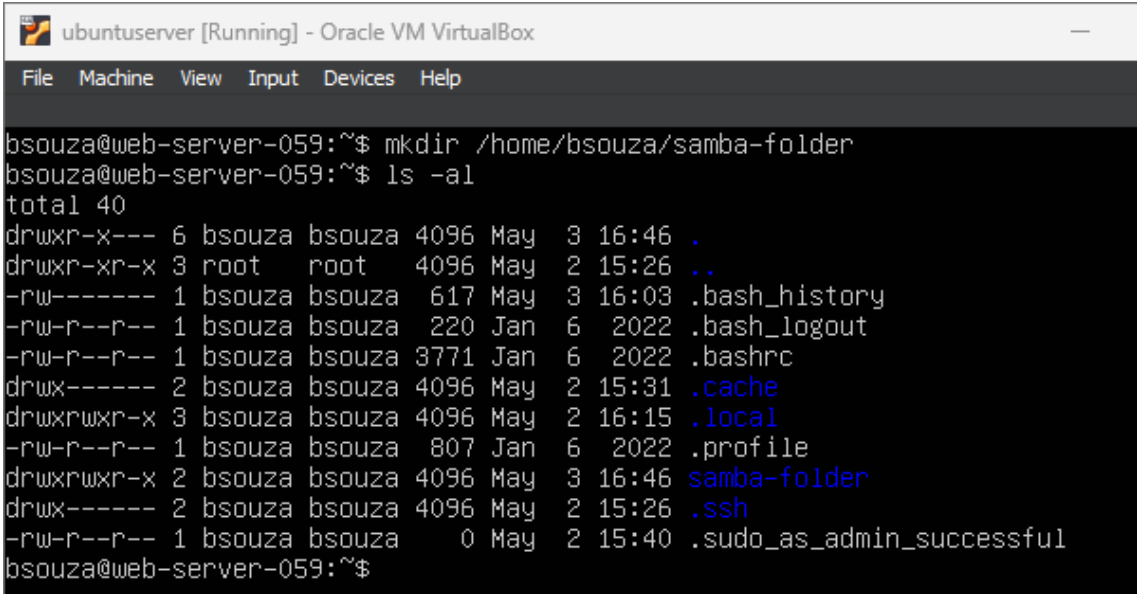
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bsouza@web-server-059:~$ sudo apt install samba -y

```

Figure 37 Installing samba server in the ubuntuserver with "sudo apt install samba -y"



```

ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-server-059:~$ mkdir /home/bsouza/samba-folder
bsouza@web-server-059:~$ ls -al
total 40
drwxr-x--- 6 bsouza bsouza 4096 May  3 16:46 .
drwxr-xr-x 3 root   root   4096 May  2 15:26 ..
-rw----- 1 bsouza bsouza  617 May  3 16:03 .bash_history
-rw-r--r-- 1 bsouza bsouza  220 Jan  6 2022 .bash_logout
-rw-r--r-- 1 bsouza bsouza 3771 Jan  6 2022 .bashrc
drwx----- 2 bsouza bsouza 4096 May  2 15:31 .cache
drwxrwxr-x 3 bsouza bsouza 4096 May  2 16:15 .local
-rw-r--r-- 1 bsouza bsouza  807 Jan  6 2022 .profile
drwxrwxr-x 2 bsouza bsouza 4096 May  3 16:46 samba-folder
drwx----- 2 bsouza bsouza 4096 May  2 15:26 .ssh
-rw-r--r-- 1 bsouza bsouza    0 May  2 15:40 .sudo_as_admin_successful
bsouza@web-server-059:~$

```

Figure 38 Creating the samba folder with "mkdir /home/bsouza/samba-folder". Then finding the Folder with "ls -al"

Figure 39 Creating a text file inside the samba folder with "nano /home/bsouza/samba-folder/my-samba-file.txt"

```
bsouza@web-server-059:~$ ls -al /home/bsouza/samba-folder
total 12
drwxrwxr-x 2 bsouza bsouza 4096 May  3 16:52 .
drwxr-x--- 6 bsouza bsouza 4096 May  3 16:46 ..
-rw-rw-r-- 1 bsouza bsouza   16 May  3 16:52 my-samba-file.txt
bsouza@web-server-059:~$
```

Figure 40 Checking the files inside the samba folder

```
GNU nano 6.2 /etc/samba/smb.conf *
; browseable = no
; create mask = 0600
; directory mask = 0700

[printers]
comment = All Printers
browseable = no
path = /var/spool/samba
printable = yes
guest ok = no
read only = yes
create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
comment = Printer Drivers
path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[samba-folder]
comment = This is my Samba setup by Barbara Souza
path = /home/bsouza/samba-folder
read only = no
browseable = yes

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line  M
```

Figure 41 Setting up the samba share

```

ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[samba-folder]
    comment = This is my Samba setup by Barbara Souza
    path = /home/bsouza/samba-folder
    read only = no
    browseable = yes

bsouza@web-server-059:~$ sudo service smbd restart
[sudo] password for bsouza:
bsouza@web-server-059:~$ sudo ufw allow samba
Rule added
Rule added (v6)
bsouza@web-server-059:~$ sudo ufw status
Status: active

To                Action            From
--                -
22/tcp            ALLOW            Anywhere
22                ALLOW            Anywhere
80                DENY            Anywhere
80/tcp            DENY            Anywhere
Samba             ALLOW            Anywhere
22/tcp (v6)       ALLOW            Anywhere (v6)
22 (v6)           ALLOW            Anywhere (v6)
80 (v6)           DENY            Anywhere (v6)
80/tcp (v6)       DENY            Anywhere (v6)
Samba (v6)        ALLOW            Anywhere (v6)

bsouza@web-server-059:~$ sudo smbpasswd -a bsouza
New SMB password:
Retype new SMB password:
Added user bsouza.
bsouza@web-server-059:~$

```

Figure 42 Allowing the Samba to share with "sudo ufw allow samba".

Setting up a password for the user with "sudo smbpasswd -a bsouza"

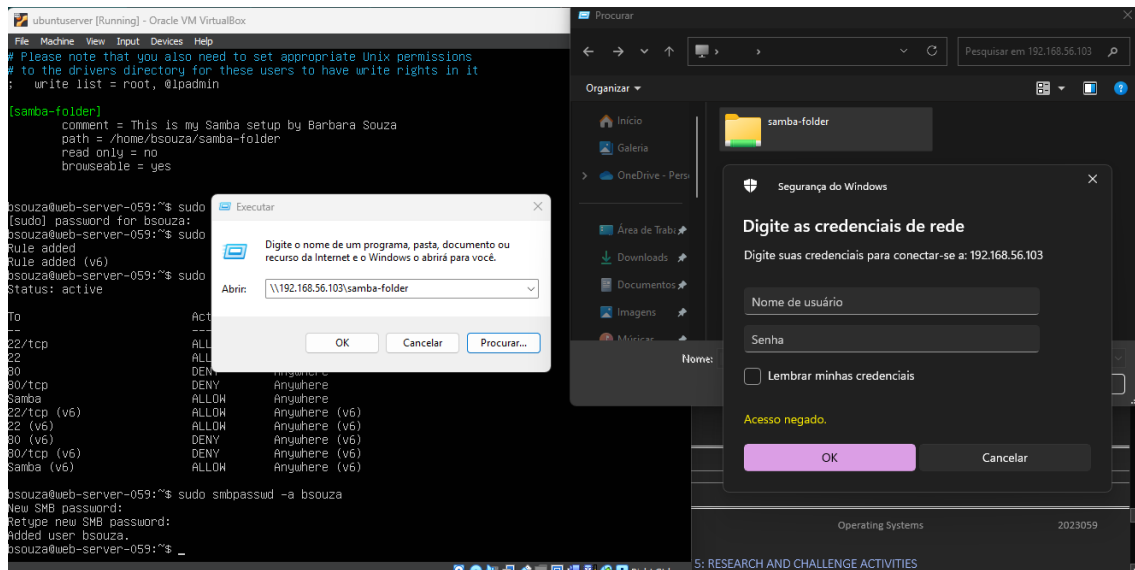


Figure 43 Accessing the file from the host computer with "\\192.168.56.103\samba-folder" then being prompted with the login with a password.

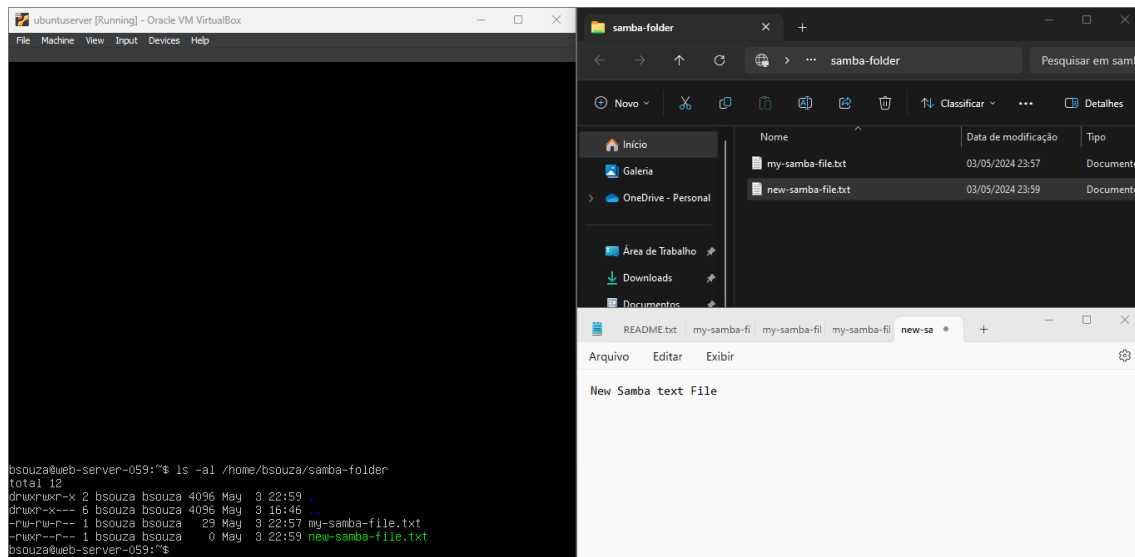


Figure 44 Adding a new text file from the host computer

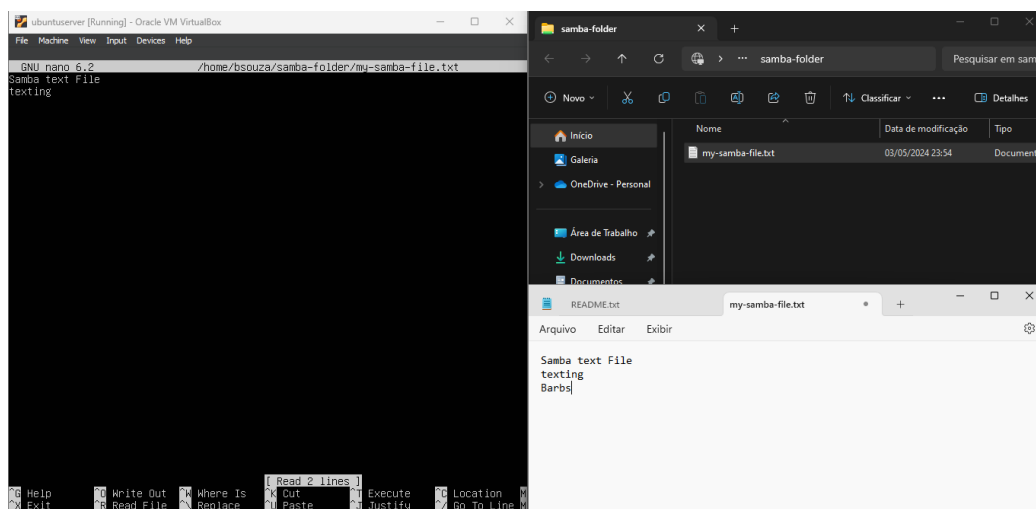


Figure 45 Editing the Text file created and shared on Linux

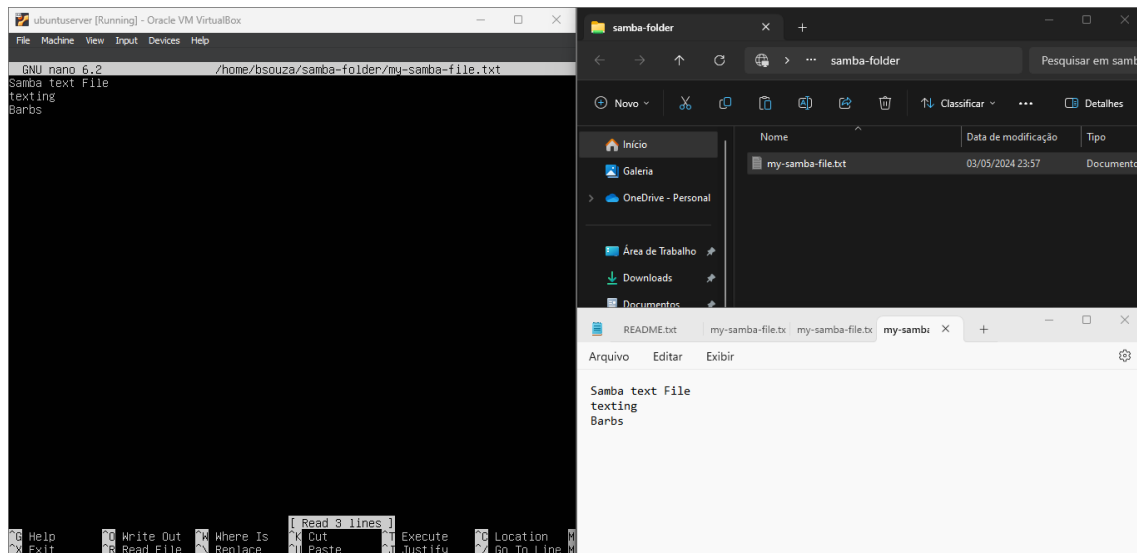


Figure 46 Text File open on Linux and on the host machine

What is SAMBA?

According to the Ubuntu website (2024) networks usually consist of various systems that don't always seamlessly integrate. To facilitate harmony between Ubuntu and Microsoft systems, a tool called Samba provides configuration options for Ubuntu Server to share network resources with Windows clients.

The Samba website (2024) says that since 1992, Samba has reliably delivered secure and rapid file and print services for all clients using the SMB/CIFS protocol, such as all versions of DOS and Windows, OS/2, Linux and many others.

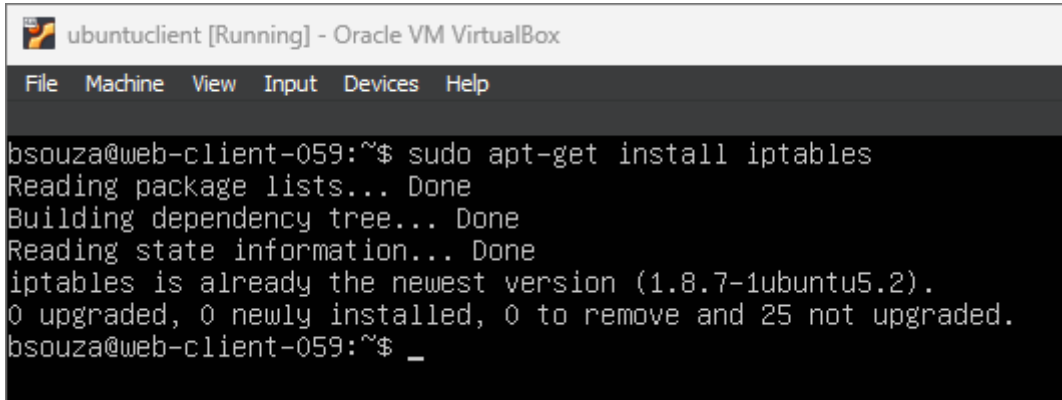
Pratama (2023) describes Samba as an open-source suite offering seamless interoperability among Windows, Linux, and Unix operating systems and desktops into Active Directory environments. It has been widely adopted solution for enabling File sharing, Print services, Authentication and Authorization, Name Resolution, Security and Integration with Windows Networking.

A Samba file server (Ubuntu, 2024) enables file sharing across distinct operating systems over a network by implementing the Server Message Block (SMB) Protocol. This protocol is a client-server communication protocol that is used for shared access to files, but also directories, printers, serial ports, and other resources on a network (ενημέρωση, 2023). The file sharing lets you access your desktop files from a laptop and share files with Windows and macOS users.

PART 5: RESEARCH AND CHALLENGE ACTIVITIES

Part 5A: Security configuration #2

This part I'll be changing the configuration of the firewall using iptables to secure the http and SSH traffic entering and exiting the server (blocking and allowing the flow). (Tucakov, 2020)

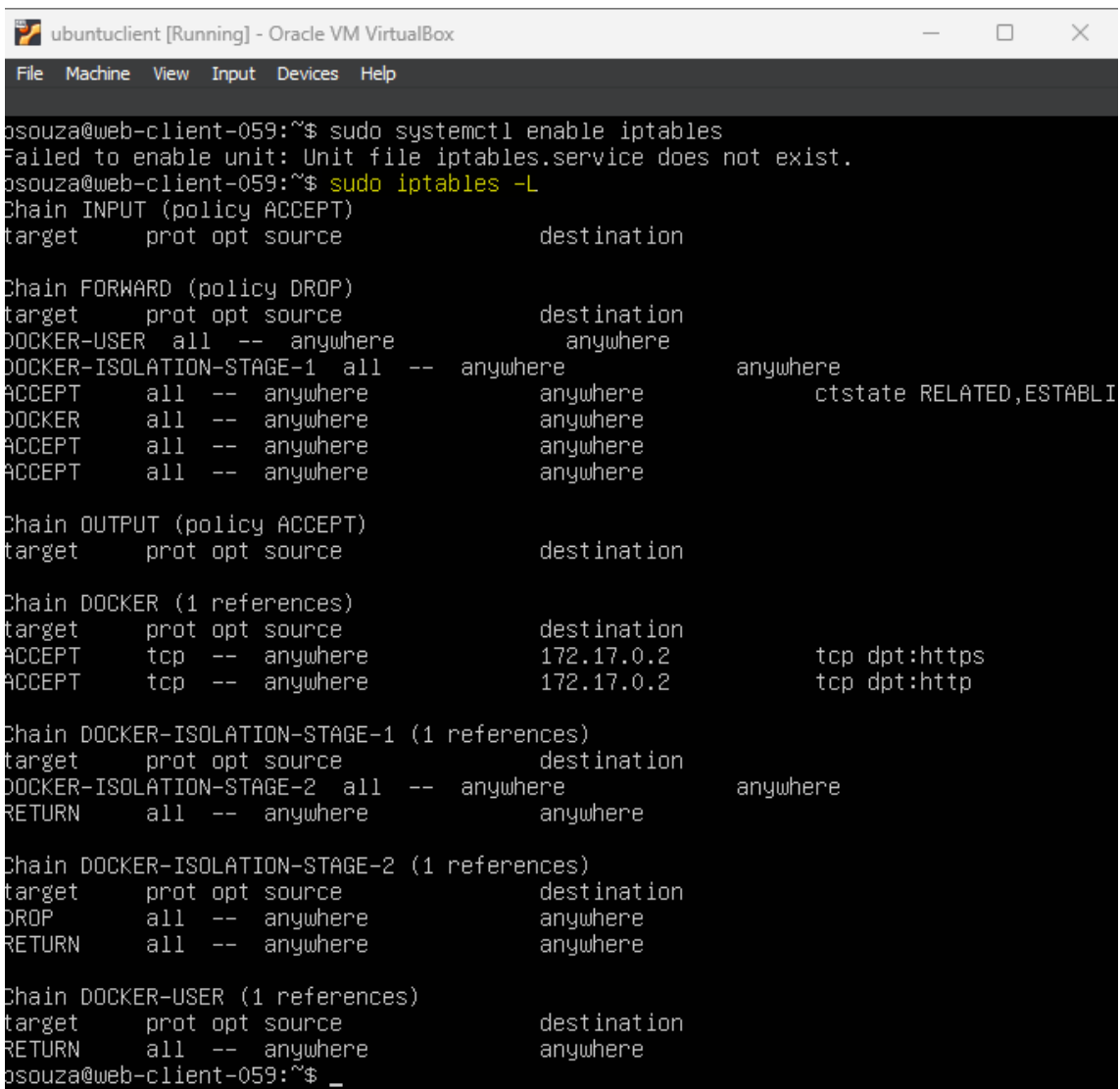


```

ubuntucient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bsouza@web-client-059:~$ sudo apt-get install iptables
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iptables is already the newest version (1.8.7-1ubuntu5.2).
0 upgraded, 0 newly installed, 0 to remove and 25 not upgraded.
bsouza@web-client-059:~$ _

```

Figure 47 Downloading iptables with "sudo apt-get install iptables"



```

ubuntucient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bsouza@web-client-059:~$ sudo systemctl enable iptables
Failed to enable unit: Unit file iptables.service does not exist.
bsouza@web-client-059:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy DROP)
target     prot opt source                destination
DOCKER-USER all -- anywhere              anywhere
DOCKER-ISOLATION-STAGE-1 all -- anywhere              anywhere
ACCEPT     all -- anywhere              anywhere             ctstate RELATED,ESTABLISHED
DOCKER     all -- anywhere              anywhere
ACCEPT     all -- anywhere              anywhere
ACCEPT     all -- anywhere              anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain DOCKER (1 references)
target     prot opt source                destination
ACCEPT     tcp -- anywhere              172.17.0.2            tcp dpt:https
ACCEPT     tcp -- anywhere              172.17.0.2            tcp dpt:http

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source                destination
DOCKER-ISOLATION-STAGE-2 all -- anywhere              anywhere
RETURN     all -- anywhere              anywhere

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source                destination
DROP       all -- anywhere              anywhere
RETURN     all -- anywhere              anywhere

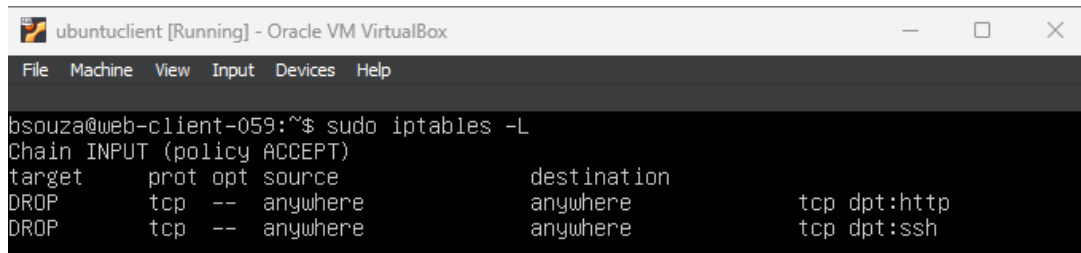
Chain DOCKER-USER (1 references)
target     prot opt source                destination
RETURN     all -- anywhere              anywhere
bsouza@web-client-059:~$ _

```

Figure 48 iptables informations with "sudo iptables -L"

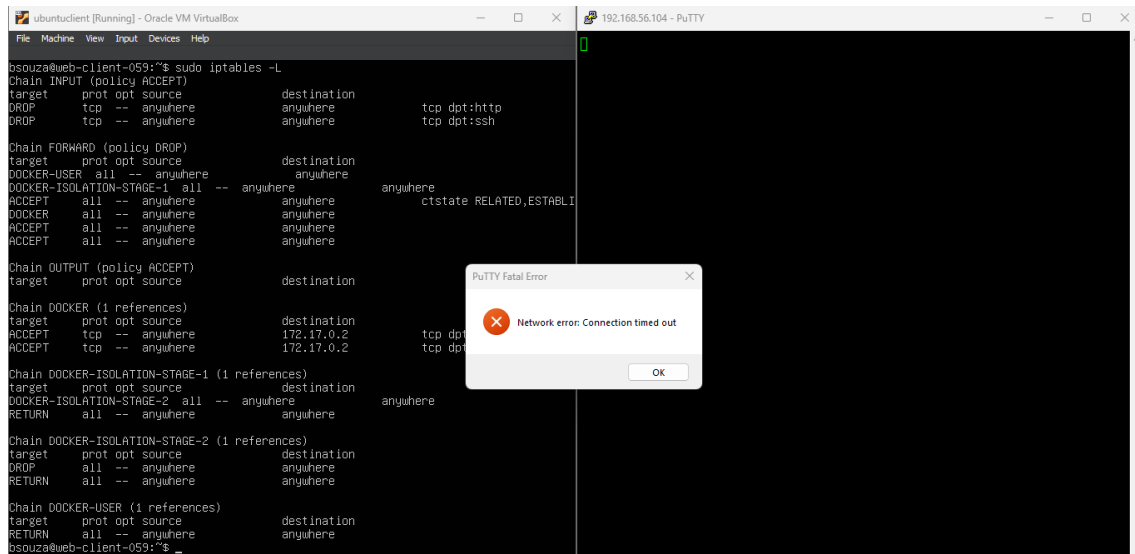
```
bsouza@web-client-059:~$ sudo iptables -A INPUT -p tcp --dport 80 -j DROP
bsouza@web-client-059:~$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
```

Figure 49 Denying the http and the SSH



```
bsouza@web-client-059:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination            tcp dpt:http
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh
```

Figure 50 Checking in iptables



```
bsouza@web-client-059:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination            tcp dpt:http
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh

Chain FORWARD (policy DROP)
target     prot opt source                destination            tcp dpt:http
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination            tcp dpt:http
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED

Chain DOCKER (1 references)
target     prot opt source                destination            tcp dpt:http
ACCEPT    tcp  --  anywhere              172.17.0.2             tcp dpt:https
ACCEPT    tcp  --  anywhere              172.17.0.2             tcp dpt:https

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source                destination            tcp dpt:http
RETURN    all  --  anywhere              anywhere               anywhere

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source                destination            tcp dpt:http
DROP      all  --  anywhere              anywhere               anywhere
RETURN    all  --  anywhere              anywhere               anywhere

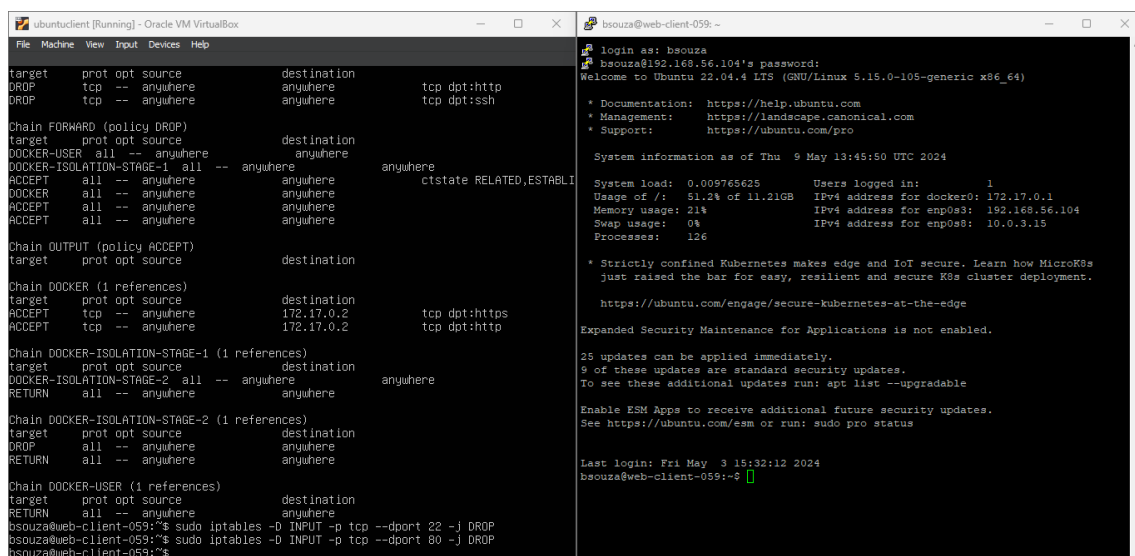
Chain DOCKER-USER (1 references)
target     prot opt source                destination            tcp dpt:http
RETURN    all  --  anywhere              anywhere               anywhere
bsouza@web-client-059:~$
```

PuTTY Fatal Error
Network error: Connection timed out

Figure 51 Not being able to access Putty

```
bsouza@web-client-059:~$ sudo iptables -D INPUT -p tcp --dport 22 -j DROP
bsouza@web-client-059:~$ sudo iptables -D INPUT -p tcp --dport 80 -j DROP
```

Figure 52 Permitting the http and the SSH



```
target     prot opt source                destination            tcp dpt:http
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh
DROP      tcp  --  anywhere              anywhere               tcp dpt:ssh

Chain FORWARD (policy DROP)
target     prot opt source                destination            tcp dpt:http
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination            tcp dpt:http
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED
ACCEPT    all  --  anywhere              anywhere               ctstate RELATED,ESTABLISHED

Chain DOCKER (1 references)
target     prot opt source                destination            tcp dpt:https
ACCEPT    tcp  --  anywhere              172.17.0.2             tcp dpt:https
ACCEPT    tcp  --  anywhere              172.17.0.2             tcp dpt:https

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
target     prot opt source                destination            tcp dpt:http
RETURN    all  --  anywhere              anywhere               anywhere

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
target     prot opt source                destination            tcp dpt:http
DROP      all  --  anywhere              anywhere               anywhere
RETURN    all  --  anywhere              anywhere               anywhere

Chain DOCKER-USER (1 references)
target     prot opt source                destination            tcp dpt:http
RETURN    all  --  anywhere              anywhere               anywhere
bsouza@web-client-059:~$ sudo iptables -D INPUT -p tcp --dport 22 -j DROP
bsouza@web-client-059:~$ sudo iptables -D INPUT -p tcp --dport 80 -j DROP
bsouza@web-client-059:~$
```

login as: bsouza
bsouza@192.168.56.104's password:
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-105-generic x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/psa>

System information as of Thu 9 May 13:45:50 UTC 2024

System load: 0.009765625 Users logged in: 1
Usage of /: 51.2% of 11.21GB IPv4 address for docker0: 172.17.0.1
Memory usage: 21% IPv4 address for enp0s3: 192.168.56.104
Swap usage: 0% IPv4 address for enp0s8: 10.0.3.15
Processes: 126

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.
<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

Expanded Security Maintenance for Applications is not enabled.

25 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

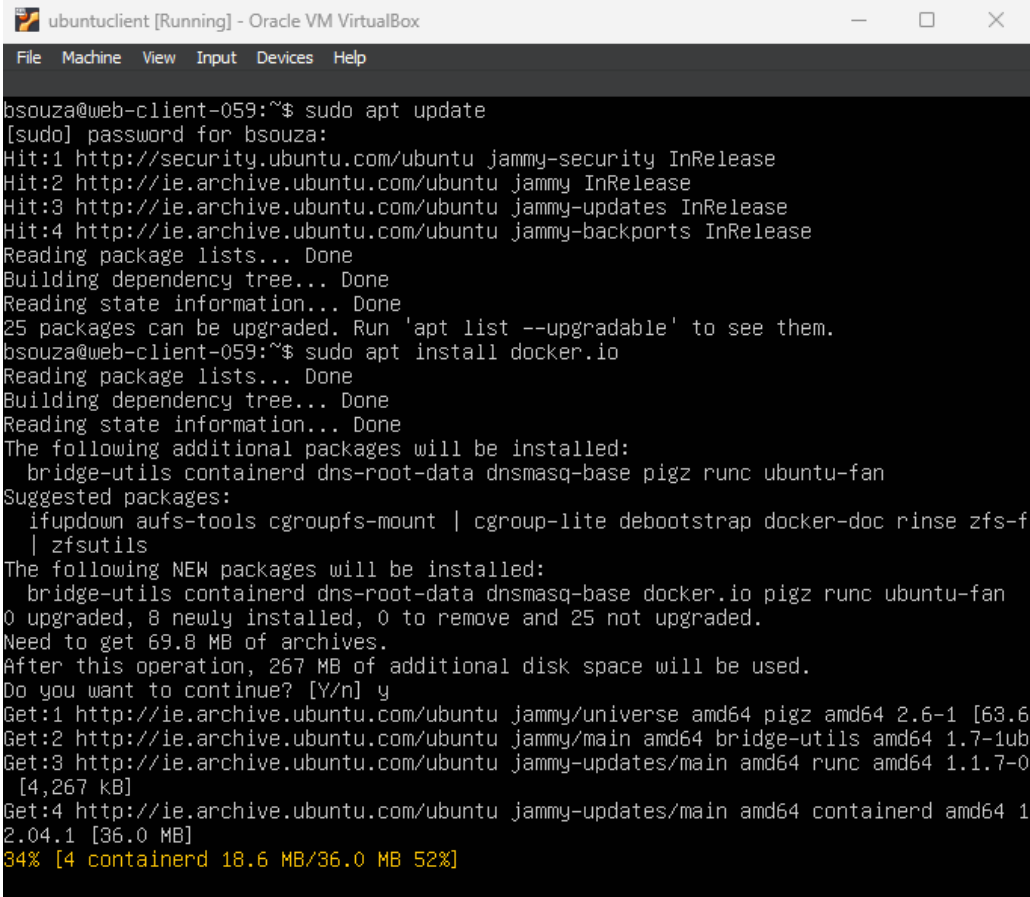
Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: sudo pro status

Last login: Fri May 3 15:32:12 2024
bsouza@web-client-059:~\$

Figure 53 Accessing Putty normally

Part 5B: Linux challenge task:

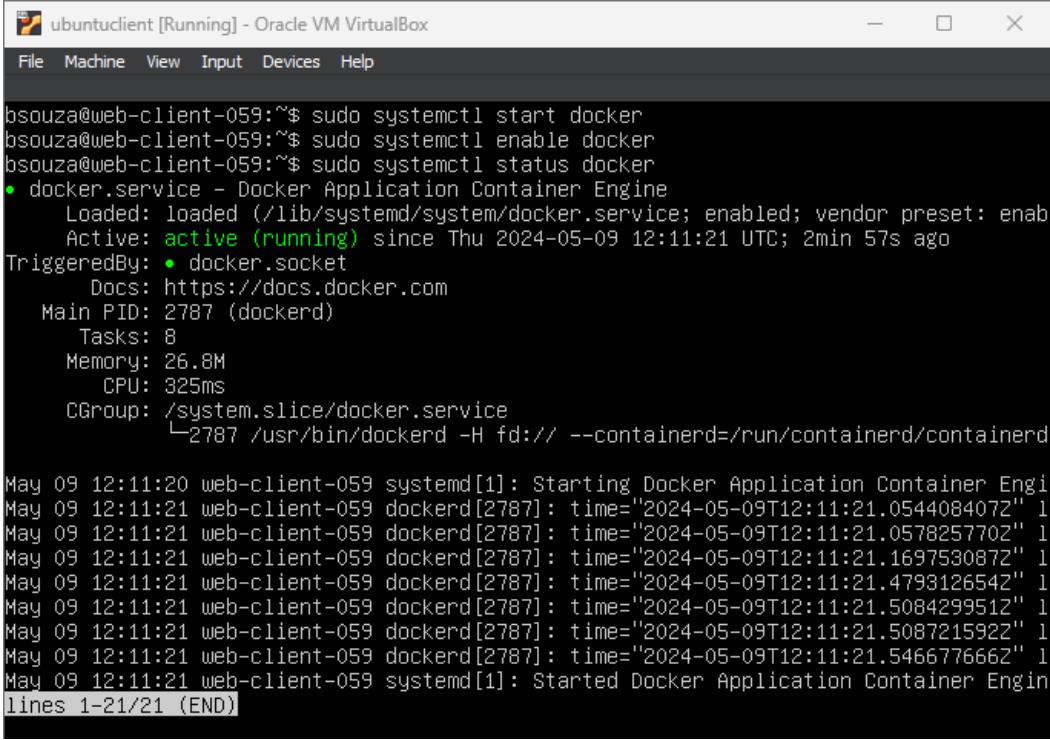
Part 5C: Container



```
ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-client-059:~$ sudo apt update
[sudo] password for bsouza:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://ie.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://ie.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://ie.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
bsouza@web-client-059:~$ sudo apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-doc rinse zfs-f
  | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 25 not upgraded.
Need to get 69.8 MB of archives.
After this operation, 267 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ie.archive.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1 [63.6
Get:2 http://ie.archive.ubuntu.com/ubuntu jammy/main amd64 bridge-utils amd64 1.7-1ub
Get:3 http://ie.archive.ubuntu.com/ubuntu jammy-updates/main amd64 runc amd64 1.1.7-0
[4,267 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu jammy-updates/main amd64 containerd amd64 1
2.04.1 [36.0 MB]
34% [4 containerd 18.6 MB/36.0 MB 52%]
```

Figure 54 "sudo apt updates" then downloading docker with "sudo apt install docker.io"



```

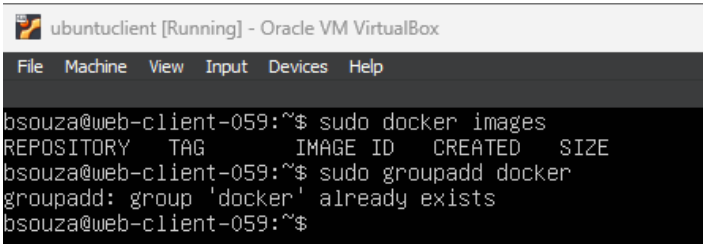
ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-client-059:~$ sudo systemctl start docker
bsouza@web-client-059:~$ sudo systemctl enable docker
bsouza@web-client-059:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enab
   Active: active (running) since Thu 2024-05-09 12:11:21 UTC; 2min 57s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2787 (dockerd)
      Tasks: 8
     Memory: 26.8M
        CPU: 325ms
    CGroup: /system.slice/docker.service
           └─2787 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd

May 09 12:11:20 web-client-059 systemd[1]: Starting Docker Application Container Engi
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.054408407Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.057825770Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.169753087Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.479312654Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.508429951Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.508721592Z" 1
May 09 12:11:21 web-client-059 dockerd[2787]: time="2024-05-09T12:11:21.546677666Z" 1
May 09 12:11:21 web-client-059 systemd[1]: Started Docker Application Container Engin
lines 1-21/21 (END)

```

Figure 55 starting and enabling docker with "sudo systemctl start docker" and "sudo systemctl enable docker". Then checking it the status with "sudo systemctl status docker" and is active.



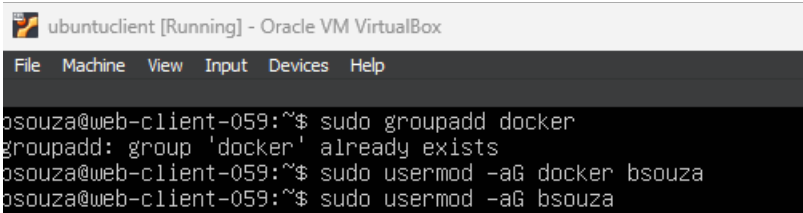
```

ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-client-059:~$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
bsouza@web-client-059:~$ sudo groupadd docker
groupadd: group 'docker' already exists
bsouza@web-client-059:~$

```

Figure 56 Checking if it is already added to the group with "sudo groupadd docker"



```

ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-client-059:~$ sudo groupadd docker
groupadd: group 'docker' already exists
bsouza@web-client-059:~$ sudo usermod -aG docker bsouza
bsouza@web-client-059:~$ sudo usermod -aG bsouza

```

Figure 57 Adding the user to the group "sudo usermod -aG docker bsouza"

```

ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bsouza@web-client-059:~$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
bsouza@web-client-059:~$ sudo groupadd docker
groupadd: group 'docker' already exists
bsouza@web-client-059:~$ sudo docker pull busybox
Using default tag: latest
^[[A^[[Alatest: Pulling from library/busybox
ec562eabd705: Pull complete
Digest: sha256:5eef5ed34e1e1ff0a4ae850395cbf665c4de6b4b83a32a0bc7bcb998e24e7bbb
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest
bsouza@web-client-059:~$ sudo docker run busybox
bsouza@web-client-059:~$ sudo docker run --rm prakhar1989/static-site
Unable to find image 'prakhar1989/static-site:latest' locally
latest: Pulling from prakhar1989/static-site
d4bce7fd68df: Pull complete
a3ed95caeb02: Pull complete
573113c4751a: Pull complete
81917632be33: Pull complete
77e66f18af1c: Pull complete
df3f108f3ade: Pull complete
d7a279eb19f5: Pull complete
e798589c05c5: Pull complete
78eeaf458ae0: Pull complete
Digest: sha256:bb6907c8db9ac4c6cadb25162a979e286575cd8b27727c08c7fbaf30988534db
Status: Downloaded newer image for prakhar1989/static-site:latest
Nginx is running...
bsouza@web-client-059:~$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
busybox latest 65ad0d468eb1 11 months ago 4.26MB
prakhar1989/static-site latest f01030e1dcf3 8 years ago 134MB
bsouza@web-client-059:~$ sudo docker run -d -P --name static-site prakhar1989/static-s
unknown flag: --name static-site
See 'docker run --help'.
bsouza@web-client-059:~$ sudo docker run -d -P --name static-site prakhar1989/static-
c95413dbc945a833428ba796e08057e402a5d643016ba56c44610dcf224cdf8
bsouza@web-client-059:~$

```

Figure 58 Downloading images with "sudo docker pull busybox", "sudo docker run busybox",

"sudo docker run --rm prakhar1989/static-site", sudo docker run -d -P --name static-site prakhar1989/static-site

Then checking with "sudo docker image"

```

bsouza@web-client-059:~$ sudo docker port static-site
80/tcp -> 0.0.0.0:32769
80/tcp -> [::]:32769
443/tcp -> 0.0.0.0:32768
443/tcp -> [::]:32768
bsouza@web-client-059:~$ sudo apt install lynx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lynx is already the newest version (2.9.0dev.10-1).
0 upgraded, 0 newly installed, 0 to remove and 25 not upgraded.
bsouza@web-client-059:~$ sudo lynx http://localhost:32769

```

Figure 59 Checking the static-site port with "sudo apt install lynx". Install Lynx with "sudo apt install lynx" and get the docker container with "sudo lynx http://localhost:32769"

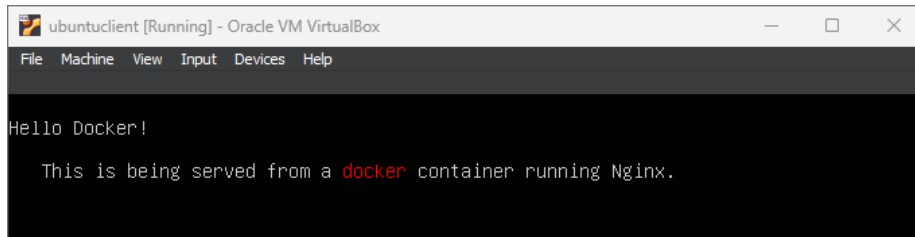


Figure 60 docker container

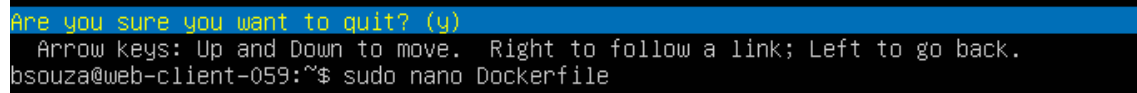


Figure 61 Creating File Dockerfile

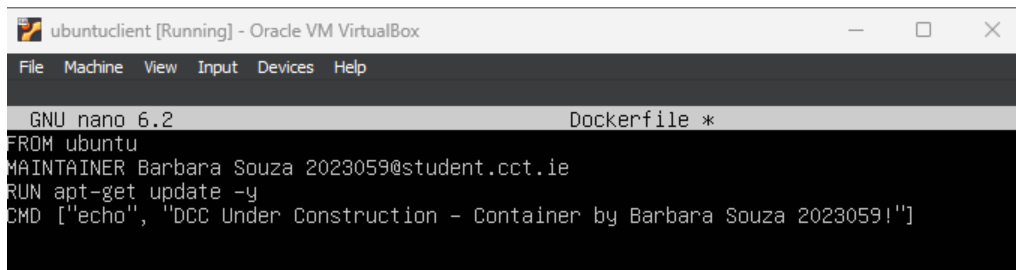
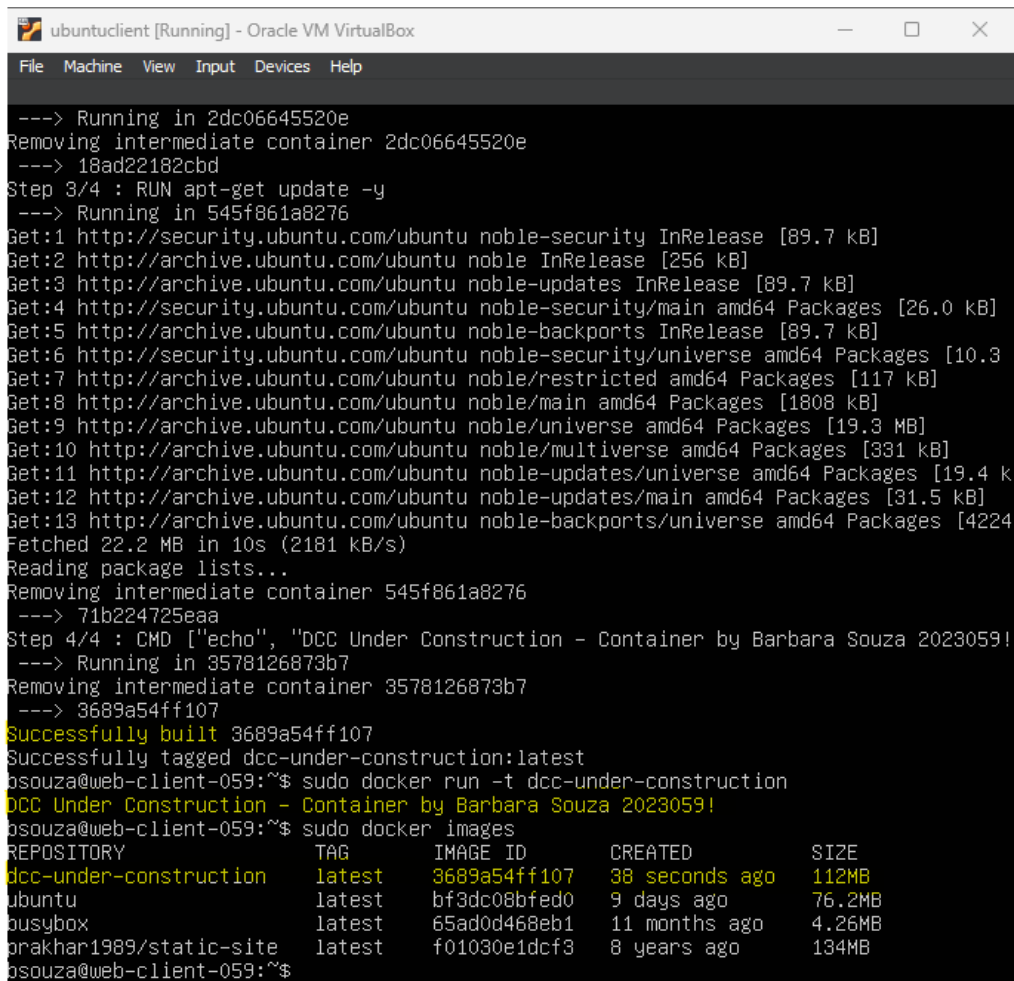


Figure 62 Creating container to show the DCC Under Construction



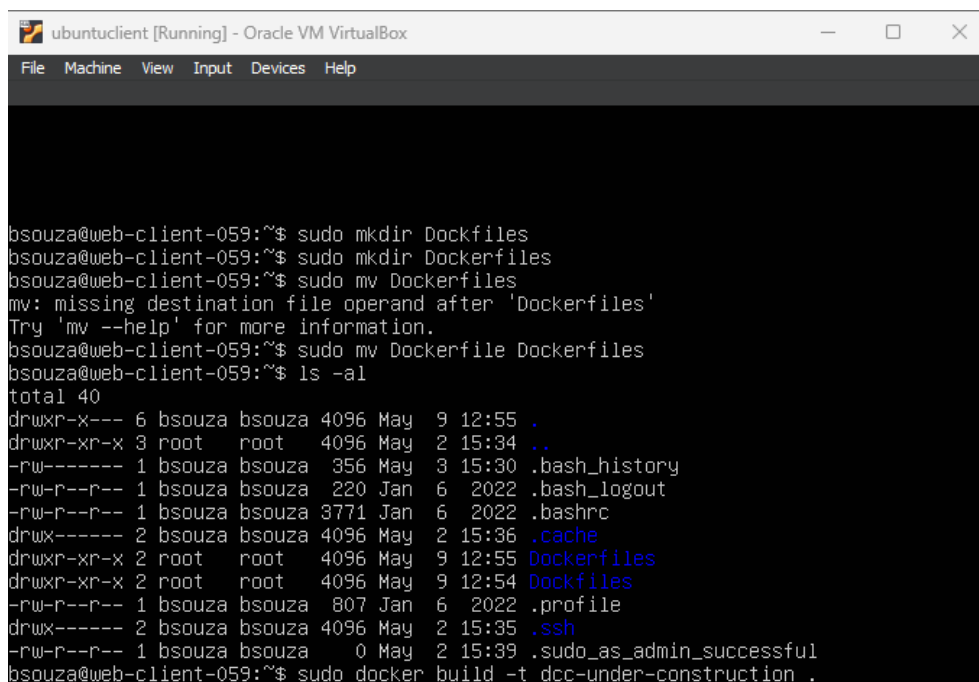
```

ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

---> Running in 2dc06645520e
Removing intermediate container 2dc06645520e
---> 18ad22182cbd
Step 3/4 : RUN apt-get update -y
---> Running in 545f861a8276
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [89.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [89.7 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [26.0 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [89.7 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [10.3
Get:7 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:10 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [19.4 k
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [31.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [4224
Fetched 22.2 MB in 10s (2181 kB/s)
Reading package lists...
Removing intermediate container 545f861a8276
---> 71b224725aaa
Step 4/4 : CMD ["echo", "DCC Under Construction - Container by Barbara Souza 2023059!"]
---> Running in 3578126873b7
Removing intermediate container 3578126873b7
---> 3689a54ff107
Successfully built 3689a54ff107
Successfully tagged dcc-under-construction:latest
bsouza@web-client-059:~$ sudo docker run -t dcc-under-construction
DCC Under Construction - Container by Barbara Souza 2023059!
bsouza@web-client-059:~$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
dcc-under-construction  latest            3689a54ff107       38 seconds ago     112MB
ubuntu               latest             bf3dc08bfed0       9 days ago         76.2MB
busybox              latest             65ad0d468eb1       11 months ago      4.26MB
prakhar1989/static-site latest             f01030e1dcf3       8 years ago        134MB
bsouza@web-client-059:~$

```

Figure 63 Creating the folder Dockerfiles "sudo mkdir Dockerfiles" and moving the file to the folder with "sudo mv Dockerfile Dockerfiles".



```

ubuntuclient [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

bsouza@web-client-059:~$ sudo mkdir Dockerfiles
bsouza@web-client-059:~$ sudo mkdir Dockerfiles
bsouza@web-client-059:~$ sudo mv Dockerfile Dockerfiles
mv: missing destination file operand after 'Dockerfiles'
Try 'mv --help' for more information.
bsouza@web-client-059:~$ sudo mv Dockerfile Dockerfiles
bsouza@web-client-059:~$ ls -al
total 40
drwxr-x--- 6 bsouza bsouza 4096 May  9 12:55 .
drwxr-xr-x 3 root   root   4096 May  2 15:34 ..
-rw----- 1 bsouza bsouza  356 May  3 15:30 .bash_history
-rw-r--r-- 1 bsouza bsouza  220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 bsouza bsouza 3771 Jan  6  2022 .bashrc
drwx----- 2 bsouza bsouza 4096 May  2 15:36 .cache
drwxr-xr-x 2 root   root   4096 May  9 12:55 Dockerfiles
drwxr-xr-x 2 root   root   4096 May  9 12:54 Dockfiles
-rw-r--r-- 1 bsouza bsouza  807 Jan  6  2022 .profile
drwx----- 2 bsouza bsouza 4096 May  2 15:35 .ssh
-rw-r--r-- 1 bsouza bsouza    0 May  2 15:39 .sudo_as_admin_successful
bsouza@web-client-059:~$ sudo docker build -t dcc-under-construction .

```

Figure 64 Then building the dcc under construction with "sudo docker run dcc-under-construction ." and run with "sudo docker run dcc-under-construction.w

Part 5D: Forgotten password

According to both Gcore (2023) and Newell (2023) to reset a lost linux password you need to follow the following steps:

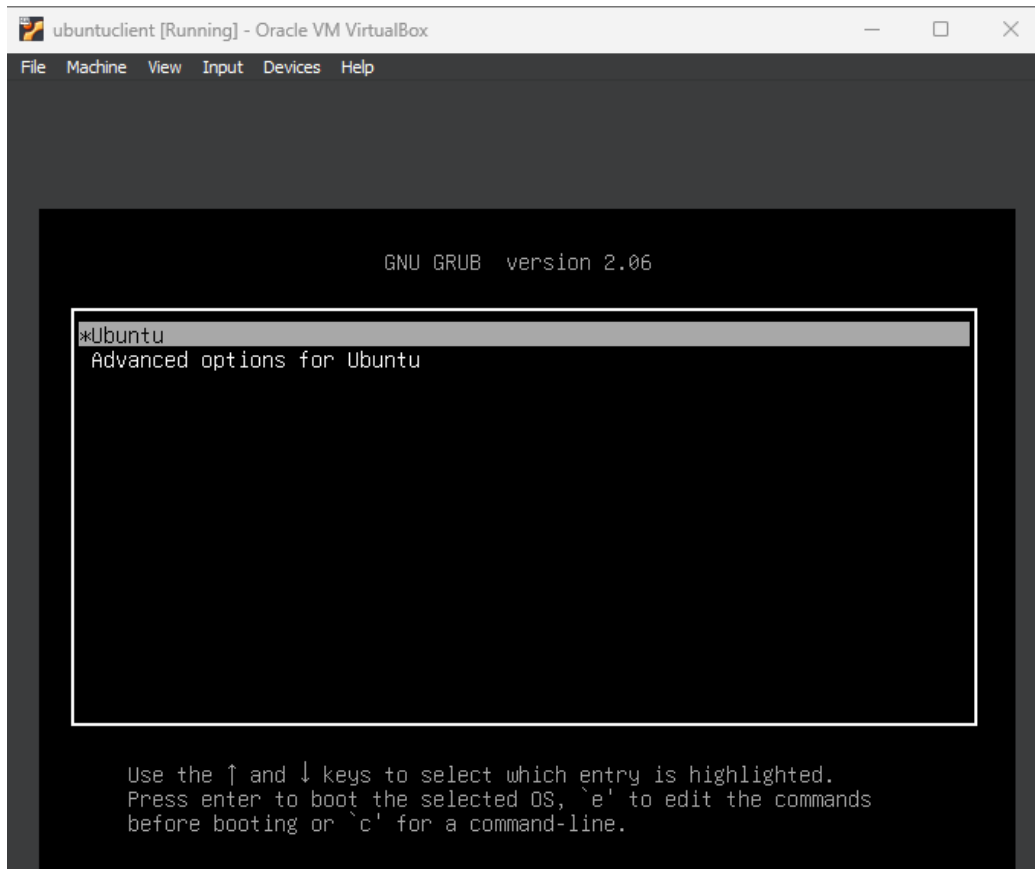


Figure 65 First when rebooting press the Esc key to go to this menu and go to Advanced options

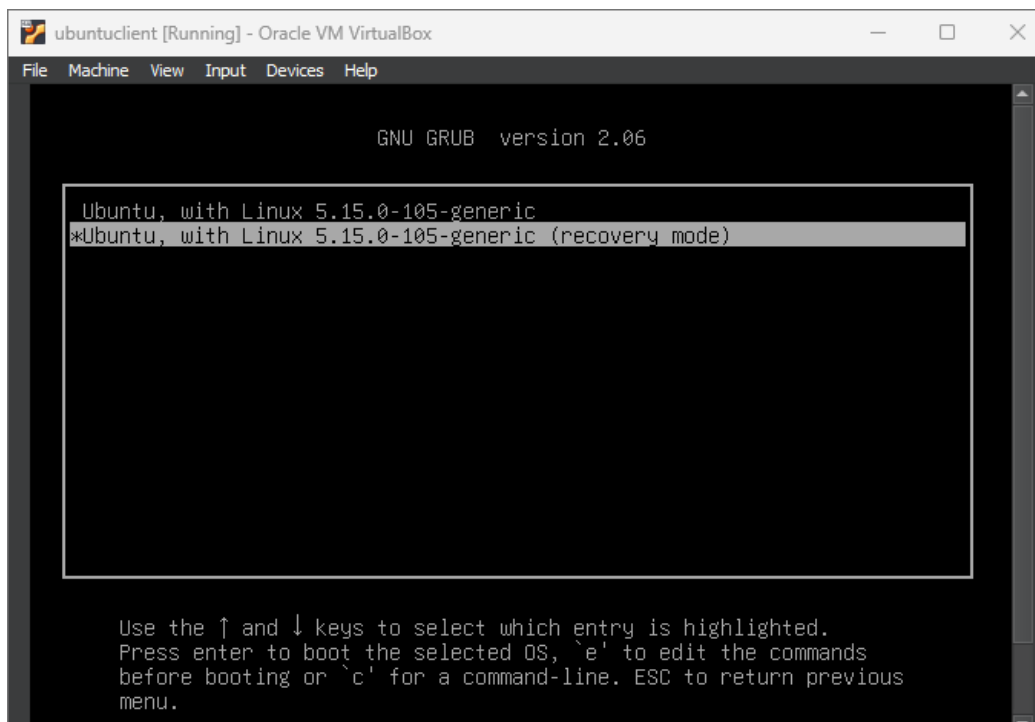


Figure 66 Select the recovery mode

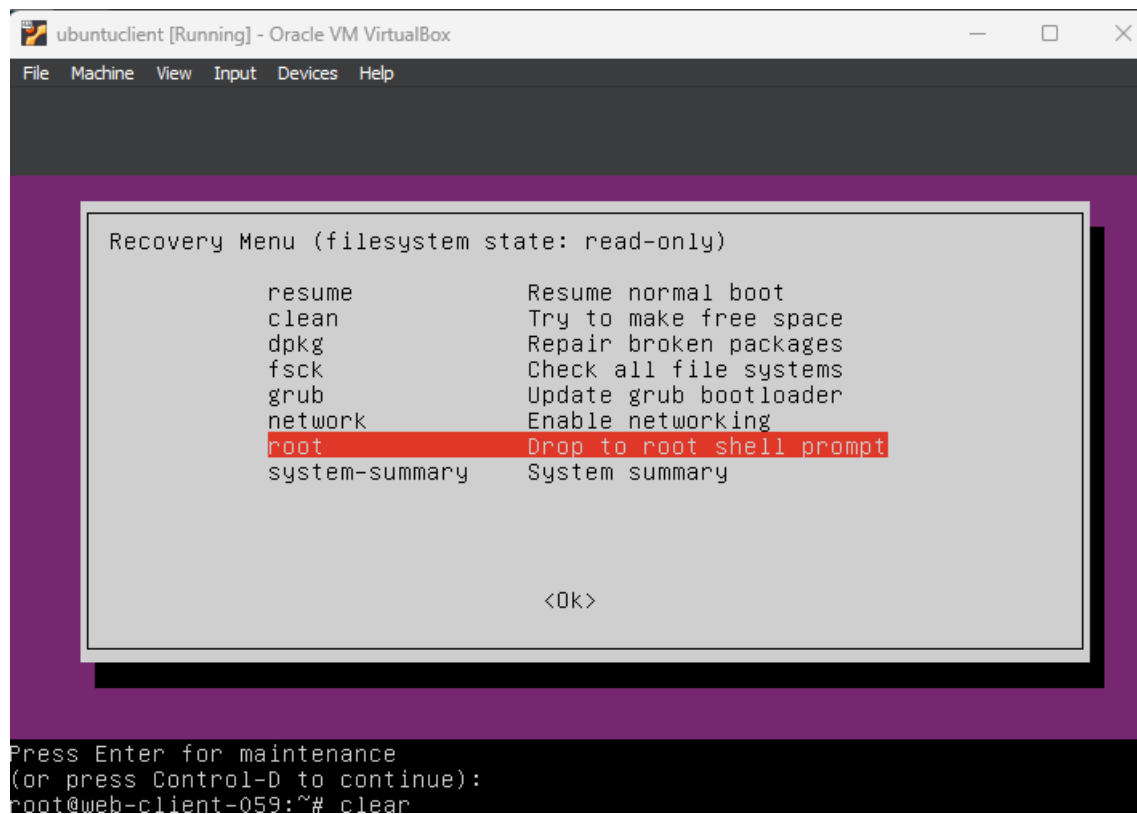


Figure 67 Then you are taken to the recovery menu, go to root - Drop to root shell prompt

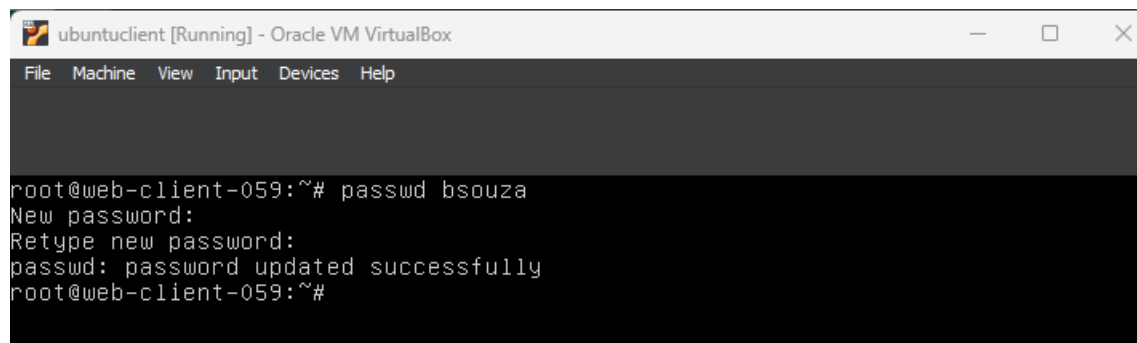


Figure 68 on the root shell prompt write passwd and your login name.

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